

Concert Table Reservation System (CTRS) for La Loy Restaurant

Milestone 1 · Part 1 GenAI Log

2110722 Software Project Management

Submitted to

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Members

SE69 Group 1 (regular program)

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GenAI Log

This log records the prompts with which the members produced the Milestone 1 artifacts of the Concert Table Reservation System (CTRS) project, and how each initial output was evaluated and revised. Each member submits two prompt log records. Every prompt is written in one of the two recommended frameworks with its components labelled: **RTF** (Role, Task, Format) for a quick request and **CREATE** (Character, Request, Examples, Adjustments, Type of output, Evaluation) for a complex task. Outputs were evaluated against the three course criteria: requirement satisfaction, PMBOK correctness and project consistency (with the CTRS Vision, Business Case, Development Case and the other artifacts of the project).

Every Milestone 1 artifact belongs, in the project's tailored RUP, to one owner role, the Project Manager (Business Case, project charter and organization are parts of the Business Case and the Software Development Plan). A one-to-one mapping of prompts to RUP roles would therefore give every prompt to one member and leave the Software Architect, the Requirements Specifier and Quality Assurance without a Milestone 1 artifact of their own. Prompt design is instead distributed as RUP distributes contributions: the owner role stays responsible for the artifact and the Review Board accepts it, while each member designs the prompts for the material their own role knows best, so that every member contributes two prompts as the course requires (Table 0.1).

Table 0.1 Prompt log records of Milestone 1

Record	Artifact	Member (prompt design)	RUP role of the member and why the material is theirs	Artifact owner (RUP)	Framework
1	Business case: background, business objective, current situation and problem	Sirapatr C.	System Analyst: owns the Vision, from which the background, the problems and the root causes are quoted	Project Manager	CREATE
2	Business case: analysis of options and recommendation	Sirapatr C.	Software Architect: assessed the alternatives and the payment gateway feasibility	Project Manager	RTF
3	Business case: critical assumptions and constraints	Chawanakorn S.	Requirements Specifier: details the constraints and quality targets in the Supplementary Specification	Project Manager	RTF

Record	Artifact	Member (prompt design)	RUP role of the member and why the material is theirs	Artifact owner (RUP)	Framework
4	Business case: preliminary project requirements	Chawanakorn S.	Requirements Specifier and User-Interface Designer: details the features into use cases, requirements and the prototype	Project Manager	CREATE
5	Business case: schedule estimate, development process and potential risks	Boonnisa T.	Project Manager: owns the iteration plans, the Development Case tailoring and the Risk List	Project Manager	CREATE
6	Project charter, stakeholder list and project organization chart	Boonnisa T.	Project Manager and Change Control Manager: owns the Software Development Plan and chairs the CCB	Project Manager	CREATE
7	Financial analysis: business model, revenue plan (plans and prices), cost model and budget estimate	Watayut A.	Quality Assurance and Test Designer: no Milestone 1 artifact of its own; brings the criteria-based, traceable evaluation the model needs (every figure tied to a stated assumption)	Project Manager	CREATE
8	Financial analysis: revenue, cost-benefit analysis, sensitivity and expansion gate	Watayut A.	Quality Assurance: independent check of the revenue assumptions and of the consistency with the Vision constraints (CON-03, CON-07)	Project Manager	RTF

Record 1: Business case, background to problem

1. Artifact	Business case Sections 2.1 to 2.3: background, business objective, current situation and problem, restated in PMBOK business-case terms from the RUP Business Case and Vision of the CTRS project.
2. Member	Sirapatr Chalermwattanatrai
3. AI Tool	ChatGPT (GPT-5, Thinking mode) at chatgpt.com; the material named in the prompt was pasted as text.
4. Initial Prompt	**[character]** You are a senior project manager in a software company preparing a business case for a small-business client under PMBOK. **[request]** Rewrite the following material of the Concert Table Reservation System (CTRS) project for La Loy Restaurant as the first three sections of a PMBOK business case, background, business objective, and current situation and problem: the problem statement, the description of the organization, the as-is tools, the ranked problem table and the project and strategic objectives of the project's Business Case and Vision (pasted with the prompt). **[examples]** Use the outline of a typical business case: 1.1 Introduction and Background (the organization, the product in one paragraph); 1.2 Project Objective (a table of SMART objectives and a table of strategic objectives with targets); 1.3 Current Situation and Problem (a table of the as-is tools, a description of the as-is process, a ranked problem table). **[adjustments]** Keep every identifier (P-nn, RC-nn, BR-nn) and every figure unchanged; do not invent volumes or numbers; write in English, concisely. **[type of output]** Markdown headings, paragraphs and tables ready to paste into the report. **[evaluation]** Check that every problem appears with its root cause and business requirement, and that nothing contradicts the Business Case and the Vision.
5. Initial Output	Three sections in the order asked for: a one-paragraph background that described La Loy and, following the wording of the pasted material, a system built "by the project team" for the restaurant at no development cost; the SMART objectives and the strategic objectives as two tables; the as-is tools table and a description of the as-is process; and the ranked problem table with impact, frequency, score, root-cause identifier and business-requirement identifier in every row. Root causes and business requirements appeared as identifiers only. Excerpt of the background as output: > La Loy is a medium-sized live-music restaurant in Bangkok that sells every table of a concert night in advance as a package. The Concert Table Reservation System (CTRS) is a LINE-integrated reservation and check-in system that the project team builds for the restaurant, so that customers reserve and pay by themselves and the Manager, Owner and Front Staff share one record. The restaurant bears no development cost; it pays only the hosting and the LINE plan, within 1,000 THB a month.

6. Review	Requirement satisfaction: background, business objective and current situation with problems present, in the outline asked for. PMBOK correctness: the business need is stated, the objectives are measurable and dated, the problems are ranked with an explicit rubric. Project consistency: every figure identical to the Business Case and the Vision. Weaknesses found: (a) the background presented the builders as a team working for the restaurant at no cost, which contradicts the sponsor and pilot customer of the charter and the investor of the financial analysis; (b) root causes and business requirements were identifiers only, so the ranking and the option comparison could not be read without the Vision; (c) the impacts of the system on the restaurant's operation (Business Case section 5) were missing, although a business case states them.
7. Revision	The prompt was re-run with the adjustments replaced and the 5 Whys, the business-requirements table and the impact table of the Business Case added to the material: **[adjustments]** The team are the engineers of one software house engaged by La Loy under a pilot implementation agreement; write the background from that position. Quote the problem table, the 5 Whys and the business-requirements table as they are, add a table that states each root cause next to its problem, and quote the impact table. Do not repeat identifiers in the prose; they stay as the keys of the quoted tables. The second output was adopted. By hand, every quoted table was replaced by an include block that regenerates from the artifact at build time, so the section cannot drift from the Vision and the Business Case.
8. Final Output	Part 2, Sections 2.1 to 2.3, as delivered below (the as-is activity diagram, Figure 2.1, is in Part 2). Acceptable because the background agrees with the parties of the charter and the financial analysis, the chain problem, root cause and business requirement can be read in the report itself, the impacts are stated, and every table is the artifact's own text. Review Board: Revise; accepted after the revision. **2.1 Background** La Loy is a medium-sized live-music restaurant and bar, open daily from 17:00 to 02:00. In the early evening it serves food and drinks; from 19:00 it serves alcoholic beverages together with a full live show. On ordinary nights it takes walk-ins and phone bookings. On nights with a concert by a well-known artist every table is sold in advance as a package: tables are grouped into Zone A (2-person round tables at 2,400 THB, 4-person square tables at 4,800 THB, 6-person sofas at 7,200 THB) and Zone B (the same three types at 2,000, 4,000 and 6,000 THB plus 1-person seats at 700 THB); each package includes a liquor tower and ice, and an extra person costs 600 THB. This advance sale is handled entirely through LINE by the Manager alone, who answers every chat and maintains the table plan by hand. The restaurant has no IT staff, and its key people (Owner, Manager, Front Staff) work evenings and nights. The four members of the team are engineers of one software house. The Concert Table Reservation System (CTRS) is that software house's product, built first for La Loy under a pilot implementation agreement and run as a tailored RUP project with versioned artifacts (Vision, Business Case, Use-Case Model, Software Requirements Specification, Development Case, iteration plans); after a pilot year with La Loy alone it is offered to other live-music restaurants as a subscription service. This business case therefore has two parties: the pilot restaurant, whose problem the product solves and whose situation, requirements and constraints Sections 2.2 to 2.6 describe, and the software house, whose investment, revenue plan and return the budget, schedule and risk sections and the financial analysis of Section 3 evaluate. The RUP Business Case of the requirements work is restated here in PMBOK terms; the figures it left qualitative are quantified.

2.2 Business Objective

The project objectives (Table 2.1) are the outcomes the project delivers; the strategic business objectives (Table 2.2) are the restaurant-level results the Owner expects from them, with the targets the Vision is held against. The baselines of the targets have not been measured yet and are to be validated with the Owner.

Table 2.1 Project objectives

No.	Objective
1	Study and develop an online concert table reservation system connected to the restaurant's LINE Official Account, through which customers can browse concert rounds, check table availability, choose their own table and pay the full table fee conveniently and in real time.
2	Reduce errors in the booking process and prevent double booking through automatic table-status management (hold with timeout, first lock wins) and automatic verification of the full table fee.
3	Develop a back-office system through which the Manager, Owner and Front Staff manage concert rounds, table layouts, prices, bookings and check-in from one shared screen and record.
4	Store booking data and customer history centrally, to notify customers at the scheduled times of their booking and to analyze artist popularity and plan the restaurant's marketing.

Table 2.2 Strategic business objectives and targets

Objective	Target	Means
Reduce the customer no-show rate	−35 %	Full table fee paid before confirmation (BRULE-01), terms accepted before paying (BRULE-16), reminders before the event and the check-in window (F32, F34)
Shorten booking and confirmation	≤ 3 minutes per booking (from hours waiting for the Manager)	Self-service seat map, 15-minute hold and gateway payment with automatic confirmation (BR-01, BR-03)
Sell every table as fast as possible on big concert nights	Table status real-time 100 %	Central database as the single record; automatic hold release; waitlist and transfer for freed tables (BR-01, BR-02)

2.3 Current Situation and Problem

2.3.1 As-is tools and process

The restaurant runs its concert-night sales on communication and note-taking tools rather than on a system (Table 2.3). The as-is process has nine steps, drawn as an activity diagram in Figure 2.1: (1) the Manager announces the concert on social media and by a LINE Official Account notification; (2) an interested customer messages the Manager on LINE to ask for free tables; (3) the Manager sends the zone-map image with the taken tables marked, editing the image by hand every time the status changes; (4) the customer chooses a table and tells the Manager its number; (5) the Manager keeps the choice in mind for the duration of the chat, normally up to an hour, without any written lock, and when several customers want the same table the first transfer slip wins; (6) the Manager sends the bank account details; (7) the customer transfers the full table fee and sends the slip; (8) the Manager checks the slip by eye, confirms the reservation and writes it into Excel or the notebook; (9) on the night, check-in opens two hours before the show and the Front Staff looks the customer up by name; arrival up to 30 minutes after the start is accepted, after that the table is released to walk-in guests and the fee is not refunded.

Table 2.3 As-is tools

Tool	Use today
LINE (Official Account and chat)	Announcements; every booking conversation; sending the zone-map image; receiving transfer slips. The Manager reads and answers each message by hand.
Excel / Google Sheets or a notebook	Record of bookings (date, name, phone, zone/table, party size, amount, status, remarks); table plan; artist schedule. No duplicate check; status updated by hand.
Facebook / Instagram	Concert schedule, artists, zone map with table types and prices.
Bank transfer and slip in chat	The customer transfers the full table fee in their own banking app and sends a screenshot; the Manager verifies it by eye and sends the account details by hand each time.

2.3.2 Problems

Six problems were identified from observation of the process, review of the restaurant's channels and the team's root-cause analysis; Table 2.4 ranks them by an Impact × Frequency rubric (each 1 to 5, team judgment, no incident data) and traces each to its root cause and to the business requirement that answers it.

Table 2.4 Problems, ranked

Rank	ID	Problem	Impact	Freq.	Score	Root cause	Business requirement
1	P-02	Double booking and data errors: table status on a hand-edited image, holds in the Manager's head, no single record; customers lose the table they picked when a faster slip arrives; disputes and refunds; confusion at the door	5	4	20	RC-02	BR-02
2	P-01	Manual booking through chat (merged with the former P-03): every step waits for the Manager; no self-service; bookings lost during booking-open waves; name lookup at check-in; missing table information; Manager hours and customer churn	4	5	20	RC-01 (RC-03 merged)	BR-01
3	P-04	Payment verification risk: slips checked by eye, bank details typed by hand; fake or wrong slips are a direct loss	4	4	16	RC-04	BR-03
4	P-06	No reminders and no terms shown before payment: forfeits and disputes at the door	3	4	12	RC-06	BR-01
5	P-05	No visibility beyond the Manager: no live status for the Owner and the staff (everything in the Manager's LINE account, a single point of failure) and no history for planning	3	3	9	RC-05	BR-04, BR-05
6	P-07	Freed tables are not resold: no waitlist when a hold expires or a no-show frees a table; no official transfer; informal resale in chat with no record	3	2	6	RC-07	BR-02

The root causes RC-01 to RC-07 named in Table 2.4 were reached by asking "why" until a structural cause that the project can act on appeared; RC-03 was merged into RC-01 when the former problem P-03 was merged into P-01.

- **P-02 double bookings.** Why do two customers get the same table? Because the map the Manager sends is already stale when the second customer picks. Why is it stale? Because the Manager edits an

image by hand and keeps pending picks in memory while chatting. Why is there no live record? Because bookings are spread over LINE, Excel and paper with no central database (RC-02).

- **P-01 manual chat process.** Why do customers wait hours? Because the Manager reads and answers every chat personally, one at a time. Why must the Manager answer? Because there is no self-service channel to see the map or reserve. Why none? Because the whole process was built on LINE chat, which has no application behind it (RC-01).
- **P-04 payment verification.** Why is verification risky? Because slips are checked by eye, one by one. Why by eye? Because nothing connects the booking to a bank or gateway. Why not? Because chat-based booking was never designed for payment integration (RC-04).
- **P-06 notifications and terms.** Why are tables forfeited without warning? Because no reminder is sent before the check-in cutoff. Why not? Because reminders and terms depend on the Manager typing them. Why is that the only way? Because no system holds the booking's timeline (RC-06).
- **P-05 visibility.** Why can the Owner not see the current bookings? Because they exist only in the Manager's chat and file. Why are they not shared? Because the tools are for communication and notes, not shared structured data (RC-05).
- **P-07 released tables.** Why are freed tables lost? Because only the Manager learns that a hold expired or a customer did not come. Why is nobody else offered the table? Because there is no queue of interested customers and no transferable booking identity (RC-07).

Table 2.5 states each root cause as the Vision records it, next to the problem it explains, so that the ranking of Table 2.4 and the comparison of the options in Section 2.5 can be read without the requirements artifacts.

Table 2.5 Root causes

ID	Root cause	Problem
RC-02	There is no single real-time record of table status; information is scattered across LINE, Excel and paper	P-02
RC-01	The whole process depends on the Manager's manual work in LINE with no system behind it, so every step waits for a human reply	P-01
RC-04	The chat-based process has no connection to a bank or payment gateway, so payment cannot be verified automatically	P-04
RC-06	Terms and reminders depend on the Manager remembering to type them; no system holds the booking timeline that could trigger them	P-06
RC-05	The tools in use (LINE, Excel, social media) are communication and note-taking tools, not a database, so data is neither shared nor analyzable	P-05
RC-07	Freed tables are known only to the Manager, there is no queue of interested customers, and a booking has no identity that can be reassigned	P-07

Each root cause is answered by one of the five business requirements BR-01 to BR-05 (Table 2.6), which are the needs the features of Section 2.6 trace to; the priority is the Vision's (1 = highest).

Table 2.6 Business requirements

ID	Need	Priority	Concerns / current solution	Proposed solution
BR-01	Automated self-service booking and customer communication: customers must be able to	1	every step waits for the Manager in chat; no	LIFF web application inside the LINE Official

ID	Need	Priority	Concerns / current solution	Proposed solution
	browse rounds, choose a zone and table, reserve and pay by themselves through LINE, and receive confirmations, terms and reminders from the system		terms or reminders (P-01, P-06)	Account with transactional and scheduled LINE messages
BR-02	Real-time availability and booking-lifecycle management: a selected table must be held and shown as taken immediately for everyone, released when the hold or grace period expires, and re-offered through a waitlist or transferred officially	1	holds in the Manager's memory, hand-edited map, first slip wins, freed tables lost (P-02, P-07)	central record with automatic hold, release, waitlist and transfer
BR-03	Integrated payment verification of the full table fee: payment must be verified automatically and the booking confirmed without human slip checking	2	bank details typed by hand, slips checked by eye (P-04)	payment gateway with automatic confirmation and e-ticket
BR-04	Centralized management dashboard: the Manager, Owner and Front Staff must manage rounds, prices and layouts, and see the same live booking status in one place	2	information locked in the Manager's chat and file; no handover (P-05, P-02)	shared real-time back-office
BR-05	Analytics and reporting: booking summaries, revenue, customer history and artist popularity must be available for planning	3	no structured data at all (P-05)	reports and export from the central record

Table 2.7 sets the proposed system against the problems from the other side: the positive and negative impacts it will have on the organization and beyond, which the options analysis of Section 2.5 and the risks of Section 2.10 take up.

Table 2.7 Impacts of the proposed system

Context	Positive impact	Negative impact
Societal	A fair first-come, first-served system; less reliance on personal connections to get a table; terms known before paying	Older customers who are less comfortable with technology may find booking harder than chatting with the Manager, so the shop must keep a staff-assisted booking path as an alternative
Environmental	Paperless: no booking notebook, no printed lists, no paper slips; all records digital	Cloud hosting and message sending add a small amount of energy use
Economic	Fewer Manager hours per booking; less revenue lost to double bookings, fake slips and unsold freed tables; reliable revenue figures per round	Initial development effort; monthly cost of hosting, APIs and the LINE Official Account plan (bounded by CON-03); a gateway fee on every payment (CON-07)
Global	Digital transformation of a restaurant business to a common international pattern; foreign customers can book through an easy, automatic flow instead of a Thai chat	Dependence on external platforms (LINE and the payment gateway) ties the system to their policies and availability

Record 2: Business case, options and recommendation

1. Artifact	Business case Section 2.5: analysis of problem resolution options and recommendation.
2. Member	Sirapatr Chalermwattanatrai
3. AI Tool	ChatGPT (GPT-5), in the same conversation as Record 1.
4. Initial Prompt	**[role]** You are a project manager writing the options analysis of a PMBOK business case. **[task]** From the four alternatives and the trade-off table of the project's Business Case (version 1.6 at the time, pasted with the prompt), write a one-paragraph recommendation that names the selected option, the reasons for it, the residual risks and the fallback option. **[format]** One paragraph of at most 120 words, followed by the two tables unchanged.
5. Initial Output	A 110-word paragraph followed by the two tables unchanged. Excerpt: > The in-house build (ALT-04) is recommended. It is the only option that answers all six root causes, keeps the restaurant's own rules (hold, grace period, transfer, package pricing) under the restaurant's control and leaves the customer and sales data with the restaurant, at no development fee and with an operating cost within the 1,000 THB a month ceiling. The residual risks are the capacity of the team and the maintenance of the system after go-live. If the build cannot be completed, the fallback is ALT-02, an off-the-shelf ticketing platform linked from the LINE Official Account, accepting its rule set and its per-ticket commission.
6. Review	Requirement satisfaction: options, comparison criteria and a justified recommendation present. PMBOK correctness: the alternatives, including a no-new-software option, are compared on the same stated criteria and the recommendation follows from them. Weaknesses found: (a) the description, cost and risk of the selected option ("no development fee", "the project team") contradicted Section 3, where the build costs the software house 767,690 THB in Year 0 and La Loy pays a pilot implementation fee; (b) the recommendation did not say under which arrangement the option is delivered; (c) the text did not state that the ratings are the team's judgment without vendor quotations.
7. Revision	The contradiction was in the source, so the Business Case itself was revised first (version 1.7: the in-house option is "a team of four engineers of one software house", its cost is the pilot implementation fee for La Loy and a product investment for the software house, its risk the team's capacity and the adoption after the pilot). The prompt was then re-run with the revised tables and the task changed: **[task]** State the recommendation as the software house's build under the pilot implementation agreement; name the residual risks and the fallback. The sentence that the ratings are team judgment without vendor quotations was added by hand.

8. Final Output

Part 2, Section 2.5, as delivered below. Acceptable because the options, their costs and their risks agree with the financial analysis and the charter, and the recommendation names the selected option, the agreement it is delivered under, the residual risks and the fallback. Review Board: Revise; accepted after the revision.

Four options were considered (Table 2.10) and compared on fit to the root causes, cost, time to operate, LINE integration, control of the restaurant's own rules, data ownership and risk (Table 2.11). Ratings are team judgment; no vendor quotation was obtained.

Table 2.10 Options considered

ID	Alternative	Description
ALT-01	Process fix without new software	Keep the LINE chat; introduce a shared Google Sheet as the single record of tables and holds, LINE reply templates for bank details and terms, and written hold and check-in rules for the Manager and the Front Staff.
ALT-02	Off-the-shelf reservation or event-ticketing platform	Subscribe to an existing restaurant-reservation or event-ticketing service and link to it from the LINE Official Account.
ALT-03	Outsourced custom build	Commission an external developer to build a system of the CTRS scope.
ALT-04 (selected)	Custom build by the team's software house: CTRS	The team, four engineers of one software house, specifies and builds a LINE-integrated concert table reservation system tailored to the restaurant's own rules, run as a tailored RUP project; the restaurant pays a fixed pilot implementation fee and the software house carries the rest of the development as its own product investment, to be offered to other restaurants after a pilot year.

Table 2.11 Comparison of the options

Criterion	ALT-01 process fix	ALT-02 off-the-shelf platform	ALT-03 outsourced build	ALT-04 CTRS (selected)
Fit to the root causes (RC-01, RC-02, RC-04..RC-07)	Partial: a single record (RC-02) but the manual bottleneck (RC-01) and the slip check (RC-04) remain	Partial: seat map and ticketing yes; hold, grace, transfer and package rules only as far as the platform allows	Full	Full
Cost	Lowest (no software)	Recurring subscription and/or per-ticket commission (not priced)	Highest: external development fee (not priced) plus operation	Fixed pilot implementation fee (150,000 THB) instead of the full price of a custom build; operation within CON-03; gateway fee per CON-07
Time to operate	Days	Weeks (set-up)	Depends on the vendor	Release 1.0 after Construction C1–C2 (April 2027)
LINE integration	None beyond the chat itself	Usually a link out of LINE to the vendor's site	Possible, at extra cost	Native: LIFF app, LINE Login, Messaging API push (CON-02, CON-10)

Criterion	ALT-01 process fix	ALT-02 off-the-shelf platform	ALT-03 outsourced build	ALT-04 CTRS (selected)
Control of the restaurant's own rules (hold, grace, transfer, pricing)	Manual, depends on the Manager following the written rules	Limited to the platform's rule set	Full, but every change costs money	Full; rules are the project's Business Rules
Data ownership	Restaurant (spreadsheet)	Platform holds customer and sales data (PDPA processor arrangement needed)	Restaurant	Restaurant (central database; CON-05)
Risk	Low technical risk; the operational risks P-01, P-02 and P-04 largely remain	Vendor lock-in; rule mismatch; commission on every ticket	Budget and vendor availability; knowledge leaves with the vendor	Team capacity; support after go-live; dependence on LINE and the payment gateway (CON-06)

Recommendation: ALT-04, the in-house custom build of CTRS. It is the only option that addresses all root causes, keeps the customer inside LINE, leaves the restaurant in control of its own hold, grace, transfer and package rules and of its data, and costs the restaurant a fixed pilot implementation fee rather than the full price of a custom build, because the software house carries the rest of the development as its own product investment. Its risks, the team's capacity, the support of the system after go-live, the dependence on the LINE Platform and the payment gateway and, for the software house, the uptake by other restaurants, are carried in Section 2.10. The process fix (ALT-01) is retained as the fallback if the project is stopped, and the bank-transfer-plus-slip path survives inside CTRS as the degraded payment mode.

Record 3: Business case, assumptions and constraints

1. Artifact	Business case Section 2.4: critical assumptions and constraints.
2. Member	Chawanakorn Sanannam
3. AI Tool	Google Gemini (Gemini 3 Pro) in the Gemini app; the two tables were pasted into the prompt as text.
4. Initial Prompt	**[role]** You are a project manager reviewing the assumptions and constraints of a software project for a PMBOK business case. **[task]** Check the assumption table (ASM-01 to ASM-08) and the constraint table (CON-01 to CON-10) of the CTRS Vision (pasted with the prompt): are they critical assumptions and constraints in the PMBOK sense, is anything missing for a project of this kind (schedule, resources, external providers), and which ones should the business case name as critical? **[format]** A short list of findings, then the two tables grouped by type with a one-sentence lead-in each.
5. Initial Output	Five findings, then the two tables grouped by type with the Rationale column of the Vision, which cites business rules and quality requirements by identifier. Excerpt of the findings: 1. The eight assumptions are conditions outside the project's control that the plan relies on, and the ten constraints are limits imposed on the solution; this is the PMBOK distinction and the tables respect it. 2. Missing: a schedule constraint for the release dates, a resource constraint on the hours the engineers can give, and an assumption that the terms of the LINE Platform and the payment gateway stay unchanged during the project. 3. Critical for the business case: the operating-cost ceiling (CON-03), the payment-fee limit (CON-07) and the dependence on external services (CON-06, ASM-02); if any of these fails, the cost side of the case changes.
6. Review	Requirement satisfaction: assumptions and constraints present, grouped, with the critical ones named. PMBOK correctness: the distinction between assumption and constraint is right, and the schedule and resource limits belong to the charter and the schedule estimate rather than to this section. Project consistency: the tables are the Vision's. Weaknesses found: (a) two of the three proposed additions repeat what the Vision already states (the external services operate under their own terms; the engineers' hours are in the schedule estimate), so only the framing was kept; (b) the Rationale column carried business-rule and quality-requirement identifiers that mean nothing in a project-management report; (c) the lead-in had to say that the operating-cost ceiling is the constraint the pricing plans of Section 3.1 must respect.

7. Revision

No prompt change. The output was edited by hand: the proposed additions were dropped, the two tables are quoted from the Vision with the Rationale column removed by the build tool, and the lead-in states what the assumptions are (conditions the plan relies on and that the team cannot control, grouped by the party they concern) and points the operating-cost ceiling to Section 3.1.

8. Final Output

Part 2, Section 2.4, as delivered below. Acceptable because the tables are the Vision's own text, grouped by type with a lead-in each and without trace identifiers, and the constraint the financial plan depends on is visible. Review Board: Accept.

The assumptions (Table 2.8) are conditions the plan relies on and that the team cannot control, grouped by the party they concern. The constraints (Table 2.9) are limits imposed from outside, grouped as business, cost, regulatory and technical. Both tables are quoted from the project's Vision, which also records the rationale of each item; the identifiers are kept as row keys.

Table 2.8 Critical assumptions

Type	ID	Assumption
Customer	ASM-01	Customers own a smartphone, already use LINE, and have a stable internet connection while reserving and paying
	ASM-03	Customers read Thai or English; the user interface is provided in Thai and English only
	ASM-04	Customers have a mobile banking application or PromptPay that can pay through the selected gateway
	ASM-08	One LINE account represents one customer; LINE Login is the customer's identity and no separate registration exists
Shop	ASM-05	The shop has at least one internet-connected computer or tablet for the back-office; Front Staff have a smartphone with a camera for QR scanning and can use a web browser and LINE
	ASM-06	The shop already has a LINE Official Account, can enable the Messaging API and LIFF, and can open a merchant account with the selected gateway after identity verification
	ASM-07	The Manager defines the round, zone map, table numbers and package prices before a round opens for booking, and the layout stays fixed for that round (no ad-hoc merging or splitting of tables)
External services	ASM-02	The external services the system depends on, the LINE Platform and the payment gateway, are highly available under their providers' service levels

Table 2.9 Constraints

Type	ID	Constraint
Business	CON-02	The customer channel must be LINE, through the LINE Official Account and LIFF; no separate mobile application
	CON-08	Package screens and LINE messages present packages factually (name, contents, price) with no wording or images that induce drinking, as required by the Alcoholic Beverage Control Act B.E. 2551

Type	ID	Constraint
Cost	CON-03	The operating cost of the system, hosting, APIs and LINE plan together, must not exceed 1,000 THB per month
	CON-07	Per-transaction payment fees, VAT included, shall not exceed 1 % of the charged amount for PromptPay QR and 3 % for cards; payment methods above these limits are not offered; the shop absorbs the fee inside the package price
Regulatory	CON-05	Personal data (name, phone, payment history) must be handled under the Personal Data Protection Act: stated purpose, consent, retention period and deletion
Technical	CON-01	The customer-facing screens must be designed mobile-first
	CON-04	Demand concentrates in the first minutes after a round opens; the design must handle concurrent reservations and table locking under burst load
	CON-06	Core functions, LINE Login, notifications and payment, depend on external APIs the team does not control; the design must tolerate their failure
	CON-09	The number of push messages of the LINE Official Account is limited by the shop's plan; notification design must stay within the quota or the budget of CON-03
	CON-10	The customer frontend must run inside LINE's in-app browser (LIFF); the back-office runs on a standard desktop browser on the shop's computer

Record 4: Business case, preliminary project requirements

1. Artifact	Business case Section 2.6: preliminary project requirements (to-be system, releases, features, quality targets).
2. Member	Chawanakorn Sanannam
3. AI Tool	Google Gemini (Gemini 3 Pro), in the same conversation as Record 3.
4. Initial Prompt	**[character]** You are a business analyst who turns a product vision into the preliminary requirements section of a PMBOK business case. **[request]** From the selected solution, the components, the release table, the feature tables (F1 to F37) and the quality targets of the CTRS Vision (pasted with the prompt), write the to-be system description, the release table, the feature tables of Release 1.0 and Release 2.0 with their traces, and a paragraph of quality targets. **[examples]** Outline of the section: a to-be description of the customer flow and the back-office; a table of the system's components; a release table (release, goal, iterations, date); one feature table per user role; the quality targets. **[adjustments]** Quote the features unchanged (ids F1 to F37, release, traces, value, cost, priority; the descriptions are in the Vision) and add a goal-per-release table; English. **[type of output]** Markdown. **[evaluation]** Every feature of the Vision appears exactly once; the release scope matches the release table.
5. Initial Output	A to-be description of the customer flow and the back-office, the components table, the release table, the four feature tables with a Traces column of problem and business-requirement identifiers, and one paragraph that listed the quality requirements by identifier with their targets. The goal-per-release table covered Release 1.0 and 2.0 only, and the feature attributes (value, cost, priority) were not explained. Excerpt of the quality paragraph: > The preliminary quality targets are NFR-01, at least 5 transactions per second at booking-open time; NFR-02, a response of the table map within 2 seconds; NFR-03, availability of 99.5 % during sale and concert hours; NFR-04, a check-in scan of under 3 seconds; and the hosting cost within CON-03.

6. Review	Requirement satisfaction: to-be system, releases and features present; the quality targets were a paragraph rather than a table. PMBOK correctness: the section states the high-level requirements and their release scope without design detail. Project consistency: the 37 features appear once each and the release scope matches the Vision. Weaknesses found: (a) the productized Release 2.1 of the pilot year (Section 2.8) had no row in the release-goal table; (b) the reader was not told what value, cost and priority mean or how the priority determines the release; (c) the Traces column and the quality-requirement identifiers repeat the requirements traceability, which this report does not need.
7. Revision	The prompt was re-run with the adjustments replaced: **[adjustments]** Add a row for Release 2.1 (product) with its goal and date; explain in the lead-in how value, cost and priority are assigned and how the priority maps to the release; present the quality targets as a table (quality requirement, target, why) without identifiers; omit the Traces column and say that the traceability is kept in the Vision. The second output was adopted, with the feature tables replaced by include blocks that quote the Vision and drop the Traces column at build time.
8. Final Output	Part 2, Section 2.6, as delivered below (the system context diagram, Figure 2.2, is in Part 2). Acceptable because every feature of the Vision appears once with its attributes explained, the three releases have a goal and a date consistent with the schedule estimate, the quality targets are a table without identifiers, and the feature text is the Vision's own. Review Board: Revise; accepted after the revision. **2.6.1 To-be system** Customers enter CTRS from the Rich Menu of the restaurant's existing LINE Official Account, sign in with LINE Login, browse the upcoming concert rounds, see a real-time table map with table types, packages and prices, select a table (which holds it for 15 minutes), accept the booking terms and pay the full table fee through the payment gateway selected by the team's feasibility study (Beam Checkout, fallback Opn Payments). The booking is confirmed automatically on the gateway's callback and a QR-coded e-ticket is sent as a LINE message. On the night, Front Staff scan the e-ticket, the booking is set to Checked-in, and tables of no-shows are released after the grace period. For the restaurant, one central database and a back-office web application replace the chat process, the hand-edited image, the slip check and the Excel record: the Manager creates rounds, zones, table types and prices, the Manager and the Owner share one live view of holds, bookings and payments, and reports support marketing and planning. Figure 2.2 shows the system context and Table 2.12 its components. **Table 2.12 Components of the to-be system** Component / Type / Description Customer Frontend / internal / mobile-first LIFF web application opened from the Rich Menu of the LINE Official Account; browse rounds, view the live table map, reserve, pay, keep the e-ticket, join the waitlist, transfer a booking Back-office Frontend / internal / web application for the Manager, Owner and Front Staff: rounds, layouts and prices, shared live dashboard, exception handling, QR scanning and check-in, reports

Component	Type	Description
Application Server	internal	cloud-hosted business logic: real-time hold and release, payment confirmation, ticket validation, notifications, waitlist, reports
Central Database	internal	encrypted relational database holding rounds, layouts, bookings, payments, tickets, customer profiles and history
LINE Platform	external	LINE Login for identity, LIFF for the web application, Messaging API for push messages, Rich Menu as the entry point
Payment Gateway	external	Beam Checkout, selected by the Payment Gateway Feasibility Study (Opn Payments as fallback): PromptPay QR, cards, mobile banking and e-wallets for the Thai market, payment result by signed webhook, refund API; per-transaction fee (CON-07)

2.6.2 Releases and features

The product is delivered in two releases (Table 2.13), each with a goal, a set of iterations and a date from the schedule estimate of Section 2.8 (Table 2.14). Tables 2.15 to 2.18 list the features by user role with the attributes the Vision records for each: its value to the restaurant (High, Medium, Low, inherited from the ranking of the problem it solves), its implementation cost as judged by the team (Low, Medium, High), and its priority 1 to 5, from which the target release follows (Release 1.0 = priorities 1 and 2, Release 2.0 = 3, Future = 4 and 5). The Vision also traces every feature to its problem and business requirement; that traceability is not repeated here. System features are automatic behavior.

Table 2.13 Releases

Release	Scope	Delivered by
1.0 Core	create a concert round, reserve and pay, watch the live booking view, check in, staff accounts, audit trail, degraded-mode switch and consent: F1, F2, F3, F4, F5, F6, F7, F8, F9, F10, F11, F12, F15, F16, F17, F21, F23, F24, F25, F27, F28; exceptions on the concert night, backup and push budget: F13, F14, F18, F19, F20, F22, F26	the Construction iterations C1 and C2 and the Transition iteration T1 (April 2027)
2.0 Growth	reminders, waitlist, transfer, reports, grace extension: F29, F30, F31, F32, F33, F34, F35	the Construction iteration C3 and the Transition iteration T2 after Release 1.0 (July 2027)
Future	F36 split payment, F37 table photos and stage distance, configurable business rules	candidates

Table 2.14 Goal and timing of each release

Release	Goal	Iterations	In operation
1.0 Core	Replace the chat process on concert nights: customers reserve and pay by themselves, every table and payment is on one record, staff check in by QR, and the Manager handles exceptions only (28 features, priorities 1 and 2)	C1, C2, T1	2 April 2027
2.0 Growth	Use the record to sell more and lose less: scheduled reminders, grace extension, waitlist, official booking transfer and reports (7 features, priority 3)	C3, T2	30 July 2027
2.1 Product	Productize the pilot system for other restaurants: multi-tenant service, Free / Standard / Premium plans and limits, subscription billing, self-service onboarding, service administration (software house scope, Section 3.1)	C4, T3	11 February 2028

Release	Goal	Iterations	In operation
Future	Candidates outside the project: split payment, table photos and stage distance, configurable business rules (priorities 4 and 5)	—	not scheduled

Table 2.15 Features for the Customer

Release	ID	Feature	Description	Value	Cost	Priority
1.0	F1	LINE sign-in	Sign in with LINE Login; no separate registration	High	Low	1
1.0	F2	Table map	View the real-time table map of a round with table status, seats, package and price	High	High	1
1.0	F3	Table selection and hold	Select a table; the system holds it for 15 minutes while paying; on the first booking enter name and phone into a profile that is pre-filled afterwards	High	Medium	1
1.0	F4	Gateway payment	Choose a payment method offered by the payment gateway (Beam Checkout) and pay the full table fee; the booking is confirmed automatically	Medium	High	1
1.0	F5	QR e-ticket	Receive a QR e-ticket for the booking	High	Low	1
1.0	F6	Transactional messages	Receive transactional LINE messages: booking confirmation, payment result and hold timeout	Medium	Low	1
1.0	F7	Booking terms	See and accept the booking terms (full payment, check-in window, grace period, no refund) before paying	Medium	Low	1
1.0	F8	My bookings	View own bookings and their status	High	Low	1
1.0	F9	Round list	Browse upcoming concert rounds with date, time and artist	High	Low	1
2.0	F29	Grace extension	Reply "on my way" or request a 30-minute grace extension from the reminder (subject to the Owner's decision on the rule)	Medium	Medium	3
2.0	F30	Waitlist	Join the waitlist of a full round or zone and be notified when a table is freed	Low	Medium	3
2.0	F31	Booking transfer	Transfer a confirmed booking officially to another LINE account	Low	Medium	3
2.0	F32	Scheduled reminders	Receive scheduled LINE reminders before the event, when the check-in window opens and before the grace cutoff	Medium	Medium	3
Future	F36	Split payment	Share a payment link so that group members pay their share of the table fee within the hold	Low	High	5
Future	F37	Table photos	See a photo of the view from each table and its distance to the stage on the table map	Low	Medium	4

Table 2.16 Features for the Manager (the Owner uses the same back-office read-only)

Release	ID	Feature	Description	Value	Cost	Priority
1.0	F10	Concert rounds	Manage concert rounds: artist, date and time, booking-open time, table layout of the round	High	Medium	1

Release	ID	Feature	Description	Value	Cost	Priority
1.0	F11	Zones and packages	Define zones, table types, packages, prices and the extra-person fee	High	Medium	1
1.0	F12	Live booking view	Live booking view: every table of the current round with its hold, booking, payment status and check-in in real time; the Owner reads the same screen read-only	High	Low	1
1.0	F13	Ticket escalation management	Resolve duplicate or invalid QR ticket cases	High	Low	2
1.0	F14	Manual booking and payment overrides	Handle exceptions: lock or unlock a table by hand, cancel a booking, confirm or reject a transfer-slip payment while the payment gateway is unreachable (degraded mode)	High	Medium	2
2.0	F33	Reports	Reports on bookings, revenue, artist popularity and customer history, with export to Excel	Low	Medium	3

Table 2.17 Features for the Front Staff

Release	ID	Feature	Description	Value	Cost	Priority
1.0	F15	QR scan	Scan the customer's QR e-ticket at the door	High	Low	1
1.0	F16	Ticket details	See the round, table number and payment status of the scanned ticket in real time	High	Low	1
1.0	F17	On-site check-in processing	Set the booking to Checked-in	High	Low	1
1.0	F18	Invalid ticket escalation	Escalate an invalid ticket to the Manager	High	Low	2
1.0	F19	No-show and table release management	Mark a no-show after the grace period; the table is released to walk-in guests or the waitlist	Low	Low	2
1.0	F20	Name lookup	Look a booking up by name or phone when the customer cannot show the QR code	High	Low	2

Table 2.18 System features (automatic behavior)

Release	ID	Feature	Description	Value	Cost	Priority
1.0	F21	Hold release	Release a held table when the 15-minute hold expires and notify the customer	High	Low	1
1.0	F22	Daily backup	Back up booking data daily and produce the Excel export	Low	Low	2
1.0	F23	Payment callback	Confirm the booking and issue the e-ticket on the gateway's payment callback	Medium	Medium	1
1.0	F24	Staff accounts	Back-office accounts with the roles Manager, Front Staff and Owner (read-only) and role-based access to each back-office function	Medium	Low	1
1.0	F25	Audit trail	Keep an audit trail of every booking state change and manual action with time, acting account and reason	Medium	Low	1
1.0	F26	Push-message budget	Count the LINE push messages per month and stop non-essential reminders within the plan's quota	Low	Low	2

Release	ID	Feature	Description	Value	Cost	Priority
1.0	F27	Degraded-mode switch	Switch new payments to degraded mode when the payment gateway does not answer and return to gateway mode when it answers again	Medium	Medium	1
1.0	F28	Consent and profile access	Obtain consent before personal data is collected and let the customer view and correct the profile	Medium	Low	1
2.0	F34	Reminder scheduling	Send scheduled reminders before the event, the check-in window and the grace cutoff	Medium	Medium	3
2.0	F35	Waitlist offer	Offer a freed table to the waitlist in queue order	Low	Medium	3

2.6.3 Quality targets

The measurable targets that the Vision hands to the Supplementary Specification bound the design and the acceptance tests (Table 2.19); the full non-functional specification is produced in Elaboration. Personal data is handled under the Personal Data Protection Act and package screens follow the Alcoholic Beverage Control Act.

Table 2.19 Preliminary quality targets

Quality requirement	Target	Why
Throughput at booking-open time	at least 5 transactions per second	demand concentrates in the first minutes after a round opens
Recovery after an outage	within 2 hours during opening hours	a concert-night sale cannot wait for the next day
Supported browsers	the current LINE in-app browser and the last two versions of Chrome, Safari and Edge	customer frontend inside LINE, back-office on the shop's desktop browser
Response time	within 2 seconds for customer and Front Staff transactions	self-service booking must not feel slower than chat
Check-in time	within 15 seconds per table	queue at the door two hours before the show
Backup	daily backup of booking data with export to Excel	the Excel record is the shop's habit and its fallback

Record 5: Business case, schedule, development process and risks

1. Artifact	Business case Sections 2.8 to 2.10: schedule estimate (phases, iterations, milestones, Gantt), development process (PMBOK process groups mapped to the tailored RUP) and potential risks.
2. Member	Boonnisa Tangjitpermkwamdee
3. AI Tool	GitHub Copilot Chat in VS Code (GPT-5); the Gantt chart was requested as PlantUML source.
4. Initial Prompt	**[character]** You are a senior project manager planning a small software project run with a tailored RUP by a team of four engineers who work on it part-time. **[request]** From the iteration contents of the project's Development Case and iteration plans, the release table of the Vision, the hours a week each of the four engineers can give per phase and the known risk seeds (pasted with the prompt), propose a schedule from 3 August 2026 with the phases Inception, Elaboration, Construction and Transition, two iterations per phase where sensible, a Release 1.0 and a Release 2.0, and the milestones LCO, LCA, IOC and Product Release; map the five PMBOK process groups onto the phases; then list the potential risks with probability, impact and a PMBOK response strategy. **[examples]** Phase table: phase, iteration, content, start, end, milestone. Risk table: id, risk, category, probability, impact, strategy (avoid, mitigate, transfer, accept; exploit or enhance for a positive risk), response. **[adjustments]** Release 1.0 must be in operation at the restaurant within the first year; include at least one positive risk. **[type of output]** A phase table with dates, a PlantUML Gantt chart, a process-group table and a risk table with ids RISK-nn. **[evaluation]** Dates consistent with the hours the four engineers can give; every risk has a response someone can own; consistency with the constraints of the Vision.
5. Initial Output	A schedule of eight iterations from 3 August 2026 to Release 2.0 on 30 July 2027 with the four milestones, a PlantUML Gantt chart, a table that assigned each PMBOK process group to a phase, a life-cycle figure that showed the requirements artifacts produced in each iteration, and fourteen risks with probability, impact, strategy and response, one of them positive. Excerpt of the process-group table:

Process group	Phase	Activities
Initiating	Inception	charter, stakeholder identification
Planning	Inception, Elaboration	Software Development Plan, iteration plans, risk list
Executing	Construction	build and test the releases
Monitoring and Controlling	Construction, Transition	iteration assessments, change control
Closing	Transition	product release, acceptance

6. Review	Requirement satisfaction: schedule, milestones, Gantt, process mapping and risks present. PMBOK correctness: the process groups were assigned to phases as if they were stages, whereas Monitoring and Controlling runs throughout and Planning repeats before every iteration; the risk responses use the PMBOK strategies. Project consistency: the iterations and the release scope match the iteration plans. Weaknesses found: (a) the schedule ended with Release 2.0, while the business model of Section 3 needs a pilot year, a productized release and the expansion decision, so the project had no Closing; (b) the development-process figure described the requirements work of each iteration rather than how the project is managed; (c) the risk list had no product-side risks (slow adoption, a competing platform, the pilot restaurant dropping out).
7. Revision	The prompt was re-run with the adjustments replaced: **[adjustments]** The plan covers Year 0 (build for La Loy, Release 1.0 and 2.0) and Year 1 (pilot operation and the productized Release 2.1), and ends with the expansion decision as the Closing; show the process groups as bars over time on a figure with the phases, the iterations and the milestones as diamonds; add the risks of the pilot and of the product. The revised schedule, Gantt chart and risk table were adopted. The process figure the tool produced was not legible at page width and was drawn by hand as an HTML figure with the same content, and a table of the project-management artifacts of each phase was added next to it.
8. Final Output	Part 2, Sections 2.8 to 2.10, as delivered below (the Gantt chart, Figure 2.3, and the development-process figure, Figure 2.4, are in Part 2). Acceptable because the schedule runs from the start of the build to the expansion decision with every milestone dated, the process groups are shown over time rather than as stages, and the fourteen risks cover the build, the pilot and the product with a strategy and a response each. Review Board: Revise; accepted after the revision.

2.8 Schedule Estimate

The project follows the four phases of the tailored RUP defined in the project's Development Case, each closed by a milestone decision: Inception (Lifecycle Objectives, LCO), Elaboration (Lifecycle Architecture, LCA), Construction (Initial Operational Capability, IOC) and Transition (Product Release). Year 0 (August 2026 to July 2027) builds Release 1.0 and 2.0 for La Loy in eight iterations; Year 1 (August 2027 to July 2028) is the pilot year, in which La Loy operates the system on its concert nights while the software house productizes it as Release 2.1 in two further iterations, and it ends with the expansion decision that closes the project (Table 2.21, Figure 2.3). In PMBOK terms the Initiating process group is I1, Planning is I2 and the two Elaboration iterations, Executing is Construction and Transition, Monitoring and Controlling runs through every iteration (weekly status assessment, iteration assessment, milestone review) and through the pilot measurement, and Closing is the expansion decision at the end of the pilot year.

Table 2.21 Schedule estimate by phase and iteration

Phase	Iteration	Content	Start	End	Milestone
Inception	I1	Problem, scope, stakeholders, high-level features; first elicitation	3 Aug 2026	24 Aug 2026	
Inception	I2	Vision revised, Development Case, Business Case, iteration plans, WBS	25 Aug 2026	2 Sep 2026	LCO, 2 Sep 2026

Phase	Iteration	Content	Start	End	Milestone
Elaboration	E1	The two main use cases (reserve a concert table, check in with e-ticket), Software Requirements Specification, Supplementary Specification	3 Sep 2026	16 Sep 2026	
Elaboration	E2	Analysis model, software architecture, user-interface prototype, test plan, Construction plan, SRS baseline	17 Sep 2026	23 Oct 2026	LCA, 23 Oct 2026
Construction	C1	Concert set-up, zones and packages, table map, table selection and hold, gateway payment, payment callback, staff accounts	26 Oct 2026	18 Dec 2026	
Construction	C2	QR e-ticket, transactional messages, check-in, live booking view, exceptions and degraded mode, audit trail, backup, consent	4 Jan 2027	26 Feb 2027	IOC, 26 Feb 2027
Transition	T1	Pilot on concert nights, training of Manager and Front Staff, data migration of the current bookings, go-live	1 Mar 2027	2 Apr 2027	Release 1.0 in operation, 2 Apr 2027
Construction	C3	Release 2.0: scheduled reminders, grace extension, waitlist, booking transfer, reports	5 Apr 2027	25 Jun 2027	
Transition	T2	Release 2.0 on La Loy's concert nights, training on the new features, end of the build year	28 Jun 2027	30 Jul 2027	Release 2.0 in operation, 30 Jul 2027
Pilot year	P1	La Loy operates Release 2.0 alone, free of charge; pilot KPIs measured on every concert night (booking time, double bookings, no-shows, check-in time, operating cost)	2 Aug 2027	28 Jul 2028	
Construction	C4	Release 2.1 productization: multi-tenant data model, plans and limits, subscription billing, self-service onboarding, service administration	2 Aug 2027	14 Jan 2028	
Transition	T3	Migration of La Loy to the productized service, release of the Free-plan sign-up, hand-over to the support routine	17 Jan 2028	11 Feb 2028	Release 2.1 in operation, 11 Feb 2028
Close-out	—	Pilot review against the charter's success criteria, expansion decision of the software house, lessons learned, project close-out	24 Jul 2028	28 Jul 2028	Expansion decision, project closed, 28 Jul 2028

2.9 Development Process

The project follows the tailored Rational Unified Process recorded in the project's Development Case rather than a sequential life cycle (Figure 2.4). Year 0 runs the four phases twice for the pilot restaurant, Inception and Elaboration once and Construction and Transition for Release 1.0 and again for Release 2.0, in eight iterations; Year 1 runs Construction and Transition a third time for the productized Release 2.1 while the pilot operation P1 measures the system on La Loy's concert nights, and ends with the expansion gate that closes the project. Every phase is closed by a milestone decision and every iteration by an assessment against its evaluation criteria. The PMBOK process groups map onto this life cycle as bars over time rather than as stages: Initiating is I1, Planning is I2 and the two Elaboration iterations and then repeats before every iteration and at every milestone, Executing is the build for the pilot restaurant, the productization and the pilot operation, Monitoring and Controlling runs through the whole project (weekly status assessment, iteration assessment, milestone

review, change control, risk review, pilot KPIs), and Closing is the expansion decision, with a phase close-out at each milestone. Table 2.22 lists the activities and the project-management artifacts of each process group, and Table 2.23 the artifacts the Project Manager produces in each phase next to the requirements and engineering work that the same iterations carry.

Table 2.22 PMBOK process groups mapped to the tailored RUP

Process group	RUP phase and iterations	Activities	Project-management artifacts
Initiating	Inception I1	Conceive New Project: problem, scope, stakeholders and high-level features agreed; first elicitation; Develop Business Case	Business Case (draft); project charter (this document); stakeholder list
Planning	Inception I2, Elaboration E1 and E2	Plan the Project: Development Case, Software Development Plan with organization and WBS, iteration plans, Risk List; Plan for Next Iteration before every iteration; use-case prioritization sets the scope of C1 and C2; financial analysis re-run at LCA	Software Development Plan; iteration plans; Risk List; scope statement, WBS, estimates, schedule and cost baseline; quality, risk and communication plans
Executing	Elaboration E1 and E2 (requirements, architecture, prototype, test plan); Construction C1, C2, C3, C4; Transition T1, T2, T3	Requirements, analysis and design, implementation, test and deployment activities of each iteration; pilot on concert nights; training; data migration	Iteration deliverables (use-case specifications, SRS, Software Architecture Document, prototype, builds, test results); Review Records
Monitoring and Controlling	Every iteration	Monitor and Control Project: weekly Status Assessment, iteration assessment against the evaluation criteria, Lifecycle Milestone Reviews (LCO, LCA, IOC, Product Release); change requests decided by the Change Control Board with impact analysis through traceability; Risk List reviewed each iteration	Status Assessments; iteration assessments; decided Change Requests; updated Risk List; earned-value and schedule tracking against the cost and schedule baselines; pilot KPI measurement
Closing	End of the pilot year	Close-Out Phase and project: pilot review against the success criteria, expansion decision of the software house, hand-over to the support routine, lessons learned, sign-off by the sponsor and the pilot customer	Pilot review; expansion decision record; hand-over record; lessons learned; final Status Assessment

Table 2.23 Project-management artifacts by phase

Phase (iterations)	Project-management artifacts	Requirements and engineering work in the same iterations
Inception (I1, I2)	Business Case with financial analysis; project charter and stakeholder list; project organization; Development Case (process tailoring); Software Development Plan (draft); iteration plans I1 and I2; Risk List (initial); LCO review	Vision, Glossary, Stakeholder Requests, Business Rules, Domain Model, Use-Case Model survey
Elaboration (E1, E2)	Scope statement; WBS and WBS dictionary; effort, resource and cost estimates; schedule (Gantt chart, critical path) and cost baseline; quality, risk and	Use-case specifications, Software Requirements Specification, Supplementary Specification, Analysis Model, Software

Phase (iterations)	Project-management artifacts	Requirements and engineering work in the same iterations
	communication management plans; iteration plans E1 and E2 and the Construction plan; LCA review with the financial analysis re-run	Architecture Document, user-interface prototype, test plan and test cases; SRS baseline under the Change Control Board
Construction and Transition (C1, C2, T1, C3, T2 in Year 0; C4, T3 in Year 1)	Iteration plan and assessment per iteration; weekly status assessments; decided change requests; updated Risk List; load-test and acceptance records; IOC and Product Release reviews; training and go-live records	Builds of the features of each iteration, tests and load test; pilot on concert nights, training, data migration, go-live; in Year 1 the multi-tenant data model, plans and limits, billing, self-service onboarding and service administration of Release 2.1
Pilot year and close-out (P1)	Monthly pilot KPI reports (booking time, double bookings, no-shows, check-in time, operating cost); plans, prices and adoption plan of the service; waiting list; expansion gate record; lessons learned; project close-out and sign-off	Operation and support of the pilot restaurant; migration of La Loy to the productized service

2.10 Potential Risks

Table 2.24 lists the risks identified at this stage with a first probability and impact rating (Low, Medium, High) and the planned response; they seed the project's Risk List, which the Project Manager reviews at the end of every iteration, and the risk register of the risk management plan. The contingency reserve of the budget (Section 2.7) is sized against them.

Table 2.24 Potential risks

ID	Risk	Category	Probability	Impact	Response
RISK-01	The restaurant-side stakeholders have not been interviewed yet and are available only in the afternoons; the Owner's approval of the Vision (LCO) and the validation of the benefit volumes depend on securing one workshop	Stakeholder	High	High	Mitigate: the System Analyst books one afternoon workshop with the Owner and Manager in September 2026; open decisions and volumes collected in a single visit; asynchronous follow-up through LINE
RISK-02	The four engineers are allocated to the project part-time; other client work of the software house delays an iteration	Resource, schedule	High	Medium	Mitigate: weekly status assessment, iteration scope re-planned at each iteration end (Manage the Scope of the System); features of priority 2 are the first to move to the next iteration
RISK-03	Merchant onboarding with the payment gateway (business type accepted, identity verification) and the LINE Official Account settings (Messaging API, LIFF) are external prerequisites that may take longer than planned or fail for the shop's business type	External	Medium	High	Mitigate: the two checks named in the feasibility study are run before LCA; Opn Payments is the fallback gateway; the degraded payment mode (bank transfer plus slip) keeps Release 1.0 operable

ID	Risk	Category	Probability	Impact	Response
RISK-04	A defect on a concert night has no responder because the same four engineers support the pilot and build the productized version at the same time	Organizational	Medium	High	Mitigate: a support routine of 8 hours a month with an on-call engineer on concert nights is budgeted from Release 1.0 (Section 3.3); the concert calendar is known weeks ahead and drives the on-call roster
RISK-05	Burst load at booking-open time exceeds the design (5 transactions a second) and holds or payments fail on the busiest nights	Technical	Medium	High	Mitigate: load test before IOC; first-lock-wins locking in the database; hosting sized to the tested load within the operating-cost ceiling
RISK-06	Outage or policy change of the LINE Platform or the payment gateway during a concert-night sale	External	Low	High	Accept with contingency: degraded payment mode, reconciliation through the gateway's status query, the Manager's manual overrides; providers' service levels monitored
RISK-07	The restaurant's operating cost exceeds the 1,000 THB a month ceiling, for example if push messages exceed the LINE plan's quota	Cost	Medium	Medium	Mitigate: push-message budget in the system, the shop's current LINE plan kept, hosting cost reviewed monthly during the pilot
RISK-08	Business rules still pending the Owner's decision (grace extension, transfer cut-off, waitlist, refund policy, retention period) change late and force rework of Release 2.0	Scope	Medium	Medium	Mitigate: decisions requested at the workshop of RISK-01 and closed before LCA; changes after the baseline go through the Change Control Board
RISK-09	Personal-data handling or the presentation of drink packages does not meet the law (Personal Data Protection Act, Alcoholic Beverage Control Act) and the shop has to stop using a function	Regulatory	Low	High	Avoid: consent, retention and correction designed in; package content guideline written in E1; QA reviews both before IOC
RISK-10	Late change requests from reviews and prototype trials widen the scope after the SRS baseline	Scope	Medium	Low	Mitigate: every request is filed as a Change Request and decided by the Change Control Board against the release plan; the requirements baseline is re-planned, not silently extended
RISK-11	(positive) The feasibility study's selected gateway offers PromptPay QR at no fee, so the gateway cost stays below the fee limit the restaurant accepts	Cost, opportunity	High	Medium	Exploit: PromptPay QR presented as the default payment method; card payments kept for foreign customers

ID	Risk	Category	Probability	Impact	Response
RISK-12	Other restaurants adopt the service more slowly than the plan of Table 3.2, and the 1,000 THB a month price ceiling leaves no room to compensate; the investment is not recovered	Market	Medium	High	Mitigate: expansion gate at the end of the pilot year (waiting list of at least 10 restaurants, La Loy on a paid plan); Free plan and the "Powered by CTRS" footer as the acquisition channel; exposure limited to the net investment of Years 0 and 1
RISK-13	A reservation or ticketing platform, or LINE itself, adds concert-table sales and undercuts the price	Market	Low	High	Accept with contingency: the shop-specific rules (hold, grace, transfer, packages) and the LINE-native flow are the differentiation; plans and prices reviewed yearly
RISK-14	The pilot restaurant stops the pilot (closes, changes Manager, loses interest) before the KPIs are measured, and the reference customer for the expansion is lost	Stakeholder	Low	High	Mitigate: pilot agreement with the free pilot year and the KPIs written down; monthly review with the Owner; a second pilot venue lined up on the waiting list

Record 6: Project charter and project organization

1. Artifact	Project charter with stakeholder list (Section 4) and project organization chart with roles and responsibilities (Section 5).
2. Member	Boonnisa Tangjitpermkwamdee
3. AI Tool	GitHub Copilot Chat in VS Code (GPT-5), same conversation as Record 5; the organization chart was first requested as PlantUML source.
4. Initial Prompt	**[character]** You are a PMO lead who writes project charters under PMBOK. **[request]** From the dates of the schedule estimate, the figures of the financial analysis, the targets of the Business Case, the stakeholder summary of the Vision and the team's RUP roles (pasted with the prompt), write the project charter of the CTRS project: title, start and finish dates, budget information, financial analysis summary, project manager, objectives, measurable success criteria, approach and stakeholder list; then propose a project organization chart that shows the sponsor, the user side, the four-member team with their RUP roles and the Change Control Board. **[examples]** Charter as a two-column table (element, content); stakeholder list as a table (stakeholder, role in the project, interest, contact); organization chart as a tree with the sponsor at the top. **[adjustments]** Use the figures of the schedule and the financial analysis exactly as given; the reviewer of an artifact is never its author. **[type of output]** A two-column charter table, a stakeholder table and a PlantUML organization chart. **[evaluation]** Success criteria measurable and traceable to the targets of the Business Case; every stakeholder of the Vision's stakeholder summary present.
5. Initial Output	A two-column charter with title, dates, budget, financial summary, project manager, objectives, six success criteria and the approach; a stakeholder table; and a PlantUML organization chart with the Owner as sponsor, the two course lecturers who had reviewed the artifacts as reviewers, the team and the CCB. Excerpt of the stakeholder table:
6. Review	Requirement satisfaction: every charter element present, success criteria measurable. PMBOK correctness: sponsor, project manager and authority stated; the CCB is shown. Project consistency: dates and budget equal to Sections 2.7 and 2.8.

Stakeholder	Role in the project	Contact
Restaurant Owner	sponsor; approves business rules and budget	La Loy
Course lecturers (2)	reviewers of the artifacts	Faculty
Boonnisa T.	Project Manager	student, Chulalongkorn University

	Weaknesses found: (a) the lecturers are not stakeholders of the software house's project and had to be removed from the list and the chart; (b) the chart had the Owner as the sponsor, while in the business model the software house sponsors the product and La Loy is the pilot customer; (c) the team members were listed with a university affiliation rather than as engineers of the software house; (d) future subscribers, who are the product's customers, were missing from the stakeholders.
7. Revision	The prompt was re-run with the adjustments replaced: **[adjustments]** The sponsor and product owner is the software house; La Loy's Owner is the pilot customer under the pilot implementation agreement; list the four members as engineers of the software house; add the future subscribers; no lecturers. Redraw the organization chart with the two parties of the agreement at the top, the project team on the sponsor side, the user side on the customer side and the CCB across both. The revised charter and stakeholder table were adopted. The PlantUML chart did not lay out legibly and was redrawn by hand as an HTML figure with the same nodes and relations.
8. Final Output	Part 2, Sections 4 and 5, as delivered below. Acceptable because the charter's parties, dates, budget and success criteria agree with the business case and the financial analysis, every stakeholder of the Vision and of the product is listed with a role and a contact, and the chart shows who sponsors, who uses and who controls change. Review Board: Revise; accepted after the revision.

4. Project Charter

Table 4.1 Project charter

Project Charter	
Project title	Concert Table Reservation System (CTRS): pilot implementation for La Loy Restaurant and productization as a subscription service
Performing organization	The software house of the four team members (company name to be confirmed by the partners)
Project start date	3 August 2026
Project finish date	28 July 2028 (Release 1.0 in operation at La Loy 2 April 2027; Release 2.0 30 July 2027; productized Release 2.1 11 February 2028; expansion decision 28 July 2028)
Budget information	Project budget 1,010,590 THB: Year 0 build 767,690 THB (labor 674,600 THB for 2,092 hours at the software house's cost rates, test devices 15,000 THB, hosting and domain 8,300 THB, contingency reserve 69,790 THB) and Year 1 pilot and productization 242,900 THB (labor 172,000 THB for 540 hours, company and legal set-up 15,000 THB, pilot hosting and support 37,200 THB, contingency reserve 18,700 THB). Revenue inside the project: La Loy's pilot implementation fee of 150,000 THB. Net investment of the software house: 860,590 THB.
Financial analysis	Software-house view over Years 0 to 5, 10 % discount rate: revenues 2,526,000 THB (pilot fee, subscriptions at 690 and 990 THB a month, setup fees; 130 paying restaurants by Year 5); NPV 18,971 THB; ROI 22 % (undiscounted); discounted payback 5.0 years; IRR 11 %; discounted life-cycle cost 1,718,913 THB. Marginal within five years, positive from Year 6; adoption is the decisive lever, so the expansion is decided at the gate at the end of the pilot year (Section 3).
Project manager	Boonnisa Tangjitpermkwamdee (Project Manager and Change Control Manager)
Project objectives	1. Deliver, for La Loy's concert nights, a LINE-integrated self-service reservation system with real-time table map, automatic 15-minute hold, full payment through a payment gateway with automatic confirmation and QR e-ticket, and a shared back-office with live booking view and QR

Project Charter	
	check-in on one central record, so that double bookings and payment fraud are eliminated, the Manager handles exceptions only, and booking data serves reminders, reports and marketing (Table 2.1). 2. Run the system through a pilot year at La Loy and measure it against the success criteria below. 3. Productize it as a multi-tenant subscription service with Free, Standard and Premium plans within the 1,000 THB a month ceiling, ready to be offered to other live-music restaurants at the expansion decision.
Project success criteria	1. Release 1.0 Core in operation at La Loy on 2 April 2027, Release 2.0 Growth on 30 July 2027 and Release 2.1 on 11 February 2028, within the project budget of 1,010,590 THB. 2. Zero double bookings on La Loy's concert nights after go-live; table status real-time on one record. 3. A booking completed within 3 minutes from table selection to confirmation, without a Manager reply (Table 2.2). 4. No-show rate at La Loy reduced by 35 % within six months of Release 2.0 (Table 2.2). 5. The restaurant's operating cost of hosting, APIs and LINE plan at or below 1,000 THB a month; check-in within 15 seconds per table; 5 transactions per second at booking-open time. 6. Vision approved by the Owner at LCO, SRS signed off at LCA, and the acceptance tests of the two main use cases (reserve a concert table, check in with e-ticket) passed with the Manager and the Front Staff. 7. At the expansion decision: La Loy continues on a paid plan, at least 10 restaurants are on the waiting list, and the per-shop hosting and support cost of the productized service is at or below the figures of Table 3.3.
Approach	1. Tailored Rational Unified Process with four phases and ten iterations (I1, I2, E1, E2, C1, C2, T1, C3, T2, then C4 and T3 in the pilot year), each closed by an assessment and each phase by a milestone decision (LCO, LCA, IOC, Product Release); the tailoring is recorded in the project's Development Case. 2. Requirements engineering with RUP artifacts (Vision, Use-Case Model, use-case specifications, Software Requirements Specification, Supplementary Specification), reviewed by an independent QA role and baselined under a Change Control Board. 3. Project management with PMBOK artifacts (business case, charter, scope statement, WBS, estimates, schedule, cost baseline, quality, risk and communication plans), produced with Generative AI as an assistant and decided by the team's Review Board. 4. Construction in two-month iterations that each deliver a testable increment, with a load test before IOC and a pilot on real concert nights in Transition. 5. Payment gateway and LINE integration verified early: sandbox from Elaboration, merchant onboarding checks before LCA, degraded payment mode as the fallback. 6. Pilot year with one restaurant before any expansion: KPIs measured on every concert night, productization in parallel, expansion decided against written gate criteria.
Stakeholders	The software house (sponsor and product owner), Restaurant Owner of La Loy (pilot customer), Restaurant Manager (customer representative, domain expert), Front Staff, customers, artists, the four-member project team, LINE Corporation, the payment gateway provider and the cloud hosting provider (Table 4.2).
Sign-off	Project Sponsor: managing partner of the software house · Pilot customer: Restaurant Owner, La Loy · Project Manager: Boonnisa Tangjitpermkwamdee · Date:

Table 4.2 Stakeholder list

No.	Stakeholder	Internal / external	Role in the project	Interest and expectation	Contact
1	The software house (its four partners as management)	Internal	Project Sponsor and product owner: funds the build and the productization, owns the product, decides the expansion at the end of the pilot year	Recover the investment through subscriptions; own a product it can sell to other restaurants	Partners' meeting
2	Restaurant Owner, La Loy	External (pilot customer)	Pilot customer executive: signs the pilot implementation agreement, approves business rules and scope, approves the Vision at LCO and signs off the SRS at LCA, decides change requests that affect scope or budget	Accurate revenue and booking figures from one record; fewer losses; operating cost within 1,000 THB a month; a fixed price for the build	In person at the restaurant, afternoons, through the Manager
3	Restaurant Manager	External (pilot customer)	Customer representative and domain expert; source of the business rules and the as-is process; sets benefit and priority of features; reviews use-case specifications and the back-office prototype; member of the Change Control Board; joins acceptance testing	Chat and data-entry workload replaced by monitoring one live view; control of rounds, prices and exceptions	LINE Official Account of the restaurant; in person, afternoons
4	Front Staff	External (pilot customer)	On-site end users; describe the check-in procedure; review and accept the check-in use case and prototype	Check-in by QR instead of name lookup; fewer seating disputes	Through the Manager
5	Customers	External	Primary users; answer the questionnaire on the LINE booking channel; a sample tries the booking prototype	Book and pay by themselves at any time; immediate confirmation, ticket and reminders	LINE Official Account; survey
6	Artists	External	Indirect stakeholder; informed through the Manager	Reliable attendance figures ahead of the show	Through the Manager
7	Boonnisa Tangjitpermkwamdee	Internal (project team)	Project Manager, Change Control Manager; Implementer in Construction	Project delivered on schedule and budget; pilot KPIs met	Software house
8	Sirapatr Chalermwattanatrai	Internal (project team)	System Analyst, Software Architect	Requirements and architecture accepted at LCO and LCA; multi-tenant architecture for Release 2.1	Software house
9	Chawanakorn Sanannam	Internal (project team)	Requirements Specifier, User-Interface Designer, Designer; Implementer in Construction	Use-case specifications, SRS and prototype accepted	Software house

No.	Stakeholder	Internal / external	Role in the project	Interest and expectation	Contact
10	Watayut Aiamanan	Internal (project team)	Quality Assurance (Requirements Reviewer, Process Engineer), Test Designer and Tester; Implementer in Construction	Reviews independent of authorship; testable requirements	Software house
11	Other live-music restaurants (future subscribers)	External (market)	Target customers of the subscription service from Year 2; a waiting list is built during the pilot year	A working, affordable concert-table system with a reference customer	Waiting list, Free-plan sign-up
12	LINE Corporation	External (platform provider)	Provides LINE Login, LIFF, Messaging API and the Official Account plan	Use within the platform's policies and quota	LINE Developers console
13	Beam Checkout (fallback: Opn Payments)	External (service provider)	Payment gateway: each restaurant's merchant account, hosted checkout, webhooks, refunds	Merchant onboarding of the shops; transaction volume	Merchant support of the gateway
14	Cloud hosting provider	External (service provider)	Hosts the multi-tenant service within the hosting assumptions of Table 3.3	Paid subscription	Provider console

5. Project Organization Chart

The project organization (Figure 5.1) has a sponsor side and a customer side. The software house, represented by its four partners as management, sponsors the project, owns the product and decides the expansion; the Project Manager leads the four-member team, in which each engineer holds several RUP roles under two rules fixed in the Development Case: every artifact has exactly one owner role, and the reviewer of an artifact is never its author, which is why Quality Assurance is a separate role that produces no requirements artifact. The Restaurant Owner of La Loy is the pilot customer under the pilot implementation agreement; the Restaurant Manager represents the users day to day and sits on the Change Control Board, with the Front Staff and the customers as end users behind the Manager. The Change Control Board (chaired by the Change Control Manager, with the System Analyst, the Manager, QA and, when scope or budget is affected, the Owner) decides change requests. Table 5.1 gives the responsibilities of each role.

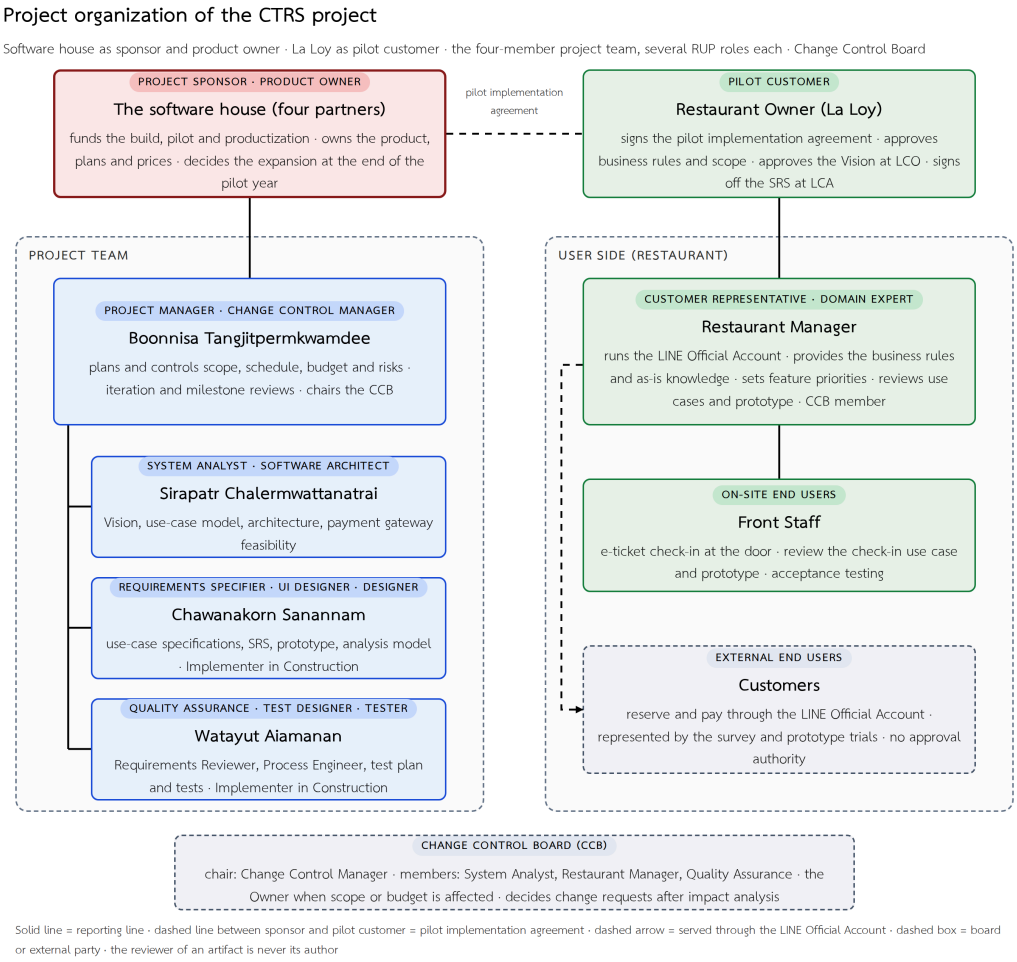


Table 5.1 Roles and responsibilities

Group	Role (holder)	Key responsibilities
Sponsor	The software house (four partners as management)	Funds the build, the pilot and the productization; owns the product and the plans and prices; reviews the pilot KPIs; decides the expansion at the end of the pilot year
Pilot customer	Restaurant Owner	Signs the pilot implementation agreement; sets business goals and key policies (zone prices, table fee, refunds); approves the Vision at LCO and signs off the SRS at LCA; decides change requests that affect scope or budget
User side	Restaurant Manager	Main customer representative; provides the business rules and as-is knowledge through interviews and observation; sets benefit and priority of features; reviews use-case specifications and the back-office prototype; Change Control Board member; joins acceptance testing
User side	Front Staff	Describe the check-in procedure and on-site problems; review the check-in use case and prototype; join acceptance testing of check-in
User side	Customers	Answer the questionnaire on the LINE booking channel; a sample tries the booking prototype; no approval authority
Project team	Project Manager and Change Control Manager (Boonnisa T.)	Plans and controls scope, schedule, budget and risks; produces the business case, charter, Software Development Plan, iteration plans, Risk List and status assessments; runs iteration assessments, milestone reviews and

Group	Role (holder)	Key responsibilities
		the pilot review; chairs the Change Control Board and keeps the baselines in Git
Project team	System Analyst and Software Architect (Sirapatr C.)	Leads elicitation and the Use-Case Model; owns the Vision, Stakeholder Requests, Glossary and Requirements Attributes; main contact with the restaurant; prioritizes use cases, produces the Software Architecture Document (including the multi-tenant architecture of Release 2.1), assesses the LINE Platform and the payment gateway
Project team	Requirements Specifier, User-Interface Designer and Designer (Chawanakorn S.)	Details the use cases and the software requirements (SRS, Supplementary Specification); models and prototypes the user interface and has users try it; produces the Analysis Model; implements in Construction
Project team	Quality Assurance, Test Designer and Tester (Watayut A.)	Reviews the requirements against the RUP checkpoints and records Review Records; produces and audits the Development Case; designs the test plan and test cases of the main use cases; runs tests, load tests and acceptance tests; implements in Construction
Board	Change Control Board	Approves, rejects or defers change requests after impact analysis through traceability; members: Change Control Manager (chair), System Analyst, Manager, QA, and the Owner when scope or budget is affected

Record 7: Financial analysis, business model, revenue plan, cost model and budget estimate

1. Artifact	Financial analysis Sections 3.1 to 3.3: business model and revenue plan (subscription plans, adoption), planning assumptions, effort and labor cost, CAPEX, OPEX, total cost, and the budget estimate of Section 2.7.
2. Member	Watayut Aiamanan
3. AI Tool	Claude Code (Claude Fable 5.1); the material named in the prompt was pasted as text, and the model was asked to keep the figures in a small script so that every table is computed from the stated assumptions.
4. Initial Prompt	**[character]** You are a project manager preparing the financial analysis of a PMBOK business case for a small software project. **[request]** From the phases with their duration and the hours a week of each engineer, the internal cost rates, the non-labor items, the operating-cost and payment-fee constraints of the Vision and the restaurant's volumes (pasted with the prompt), draft the cost side of the financial analysis of the CTRS project: the development effort and labor cost per phase, the capital expenditure, the operating cost per year, the total cost, and the budget estimate with a contingency reserve. **[examples]** Tables of a typical business case: effort per phase (hours and cost per role), CAPEX (item, unit price, units, total), OPEX per year (item, Year 0 to Year 5, total), total cost, budget (item, basis, THB). **[adjustments]** Every figure must follow from a stated assumption; keep the operating cost within the 1,000 THB a month ceiling; use the payment-fee limits given; no management reserve; THB. **[type of output]** Markdown tables with a one-line basis for each figure, and a note of every assumption made. **[evaluation]** Totals add up; the categories are the PMBOK ones (CAPEX, OPEX, contingency reserve, total cost); nothing contradicts the constraints.
5. Initial Output	A cost model as a small Python script so that every figure in the tables is computed from named assumptions: effort per phase from the hours a week of the four members at 300 to 350 THB an hour (2,092 hours, 674,600 THB); CAPEX limited to door equipment (11,500 THB) because the system runs inside LINE and a browser; Year 0 non-labor cost 8,300 THB; operating cost 68,870 THB a year (hosting within the ceiling, a maintenance retainer, gateway fees at 0.6 % of the table fees processed); a 10 % contingency reserve; Year 0 budget 763,840 THB. The model was written from the restaurant's side: the team's labor was a cost of the project, the operating cost was the restaurant's, and the gateway fee was computed on the restaurant's table fees. Excerpt:

Item	Basis	THB
Development labor	2,092 hours at the internal cost rates	674,600
CAPEX	door equipment	11,500

Item	Basis	THB
Non-labor development cost	hosting, domain, certificate	8,300
Contingency reserve	10 %	69,440
Project budget, Year 0		763,840

6. Review

Requirement satisfaction: all cost categories present (effort, CAPEX, OPEX, total cost, budget). PMBOK correctness: a contingency reserve is sized against identified risks and a management reserve is stated as not budgeted. Project consistency: hosting stays under the ceiling; the gateway rates are those of the feasibility study.

Weaknesses found at the review: (a) the analysis mixed two parties, it charged the team's labor as a cost but the operating cost and the gateway fees were the restaurant's, so it answered neither the restaurant's nor the developer's question; (b) the restaurant's benefits were about to be credited as the return (Record 8), which the software house never receives; (c) there was no revenue side of the software house's own, so the case could not say why the software house should build the product.

7. Revision

Revised prompt (RTF):

[role] You are the managing partner of a four-person software house.

[task] Re-frame the CTRS financial analysis from the software house's side: the restaurant pays a fixed pilot implementation fee, the software house owns the product, runs a one-restaurant pilot year while productizing it, and then sells it to other live-music restaurants as a subscription at no more than 1,000 THB a month; design Free, Standard and Premium plans with limits, a setup fee and an adoption plan; add the pilot year's effort, hosting and legal set-up to the costs; drop the restaurant's benefits from the return.

[format] The same tables as before plus a plan table and an adoption table; every figure computed from a named assumption.

The output was adopted with hand edits: the plan limits were aligned with the feature list, hosting and support were scaled per shop, and the restaurant's benefits were reduced to one sentence of pricing rationale. The hours a week, the cost rates and the plan prices were confirmed at the team review.

8. Final Output

Part 2, Sections 2.7 and 3.1 to 3.3, as delivered below. Acceptable because the analysis now has one investor (the software house) and one revenue source (what restaurants pay), every figure is traceable to a stated assumption, the categories follow PMBOK (CAPEX, OPEX, contingency reserves, total cost) and the prices respect the operating-cost ceiling and the payment-fee limits. Review Board: Revise; accepted after the revision.

2.7 Budget Estimate

The budget is the software house's cost of the project: the effort of the four engineers per phase at the software house's internal cost rates, the test devices, the hosting and services of the build and the pilot, the company and legal set-up of the subscription business in Year 1, and a contingency reserve (Table 2.20). Year 0 builds Release 1.0 and 2.0 for the pilot restaurant; Year 1 runs the pilot and productizes the service (Release 2.1). The only revenue inside the project is La Loy's pilot implementation fee; the subscription revenues start after the expansion decision and are analyzed in Section 3.

Table 2.20 Budget estimate

Category	Basis	THB
CAPEX, Year 0	Test devices (Table 3.5)	15,000
Development labor, Year 0	2,092 hours of the four engineers at 300 to 350 THB/h, Release 1.0 and 2.0 for the pilot restaurant (Table 3.4)	674,600
Hosting and services, Year 0	Development, staging and pilot hosting, domain (Table 3.6)	8,300
Contingency reserve, Year 0	10 % of the Year 0 cost, for the risks of Table 2.24	69,790
Budget, Year 0 (build)		**767,690**
Development labor, Year 1	540 hours: multi-tenant productization, plans and billing (C4, T3)	172,000
Company, legal, hosting and support, Year 1	Registration and legal documents, pilot hosting and support (Table 3.6)	52,200
Contingency reserve, Year 1	10 % of the productization work	18,700
Budget, Year 1 (pilot and productization)		**242,900**
Project budget, Years 0 and 1		**1,010,590**
Revenue inside the project	Pilot implementation fee paid by La Loy in Year 0	150,000
Net investment of the software house	budget less the pilot fee	**860,590**

3.1 Business Model and Revenue Plan

The four members of the team are engineers of one software house, and CTRS is that software house's product. La Loy is the pilot customer: it pays a fixed pilot implementation fee for the build of Release 1.0 and 2.0 (Year 0) and uses the system free of charge during the pilot year (Year 1), while the software house productizes it (multi-tenant service, plans and limits, subscription billing, self-service onboarding, Release 2.1) and measures the pilot against the success criteria of the charter. At the end of the pilot year the software house decides whether to expand; from Year 2 the service is offered to other live-music restaurants and bars that sell concert tables in advance, as a subscription.

The pricing follows the restaurant's own constraint: a restaurant of this size will pay at most 1,000 THB a month for hosting, APIs and the LINE plan together (constraint CON-03 of Table 2.9), so the top plan costs 990 THB a month with hosting included, the restaurant keeps its own LINE Official Account plan and its own merchant account with the payment gateway (the transaction fees stay with the restaurant), and the software house carries no payment-processing cost. Three plans (Table 3.1) separate acquisition from revenue: the Free plan lets a restaurant run one concert round a month with the core flow and the "Powered by CTRS" footer on the customer screens, which is the product's own marketing channel; Standard sells every Release 1.0 feature; Premium adds the Release 2.0 features that earn the restaurant money on top of the core flow (reminders, waitlist, transfer, reports). Every paid subscription starts with a setup fee that covers the configuration and training of the shop. For the restaurant the Premium fee of 11,880 THB a year is small against the Manager time and the lost or unsold tables it replaces (one lost table a month already costs more than the fee); those benefits are the customer's reason to subscribe and are deliberately not counted in the software house's analysis below.

Table 3.1 Subscription plans and fees

Plan	Price (THB a month, VAT excluded)	Contents and limits
Free	0	1 concert round a month; up to 20 tables; Release 1.0 core: round list, table map, hold, gateway payment through the shop's own merchant account, QR e-ticket, check-in, name lookup; 1 Manager and 1 Front Staff account; "Powered by CTRS" on the customer screens; self-service set-up; community support (LINE group)
Standard	690	Unlimited rounds; up to 60 tables; every Release 1.0 feature: live booking view, exceptions and degraded payment mode, audit trail, daily backup with Excel export; up to 5 staff accounts; support by LINE within one business day
Premium	990	Standard plus the Release 2.0 features: scheduled reminders, grace extension, waitlist, booking transfer, reports and analytics; unlimited tables and staff accounts; concert-night priority support (response within 2 hours)
Setup (paid plans, one-time)	2,500	zone map, packages and prices configured; LINE Official Account and payment gateway connected; 2-hour training of the Manager and Front Staff
Pilot implementation (La Loy, Year 0, one-time)	150,000	custom build of Release 1.0 and 2.0 for the pilot restaurant, paid 30 % at LCA, 40 % at IOC and 30 % at go-live; Premium free of charge during the pilot year

The adoption plan (Table 3.2) assumes an addressable market of about 500 live-music venues in Bangkok and the major cities that sell tables for concert nights, of which 130 pay for the service by the end of Year 5 (26 %), with a Free-plan base about as large as the paying base and a churn of 10 % a year. The plan is deliberately slow in Year 2, when the software house sells with one reference customer.

Table 3.2 Adoption plan

Shops	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Paying shops at year end (Standard + Premium)	-	-	15	40	80	130
Free-plan shops at year end	-	La Loy (pilot)	20	50	90	140
New paid subscriptions during the year (incl. replacement of 10 % churn)	-	-	15	27	44	58
Average paying shops during the year	-	-	8	28	60	105

3.2 Basis of the Analysis

The analysis covers Year 0, the build year from 3 August 2026 to 30 July 2027, Year 1, the pilot and productization year to 28 July 2028, and four expansion years, Years 2 to 5 (August 2028 to July 2032). Costs and revenues are in 2026 Thai Baht, held constant, VAT excluded (the software house stays below the VAT registration threshold within the horizon), and discounted at 10 % a year. The software house's labor is costed at its internal cost rates although the four partners could choose to defer their own pay; the partners' cash view is shown separately in Section 3.6. The restaurant's benefits are not part of the analysis. The volumes that drive the revenue, the number of restaurants that adopt the service and the plan mix, are planning assumptions (Table 3.3) that the pilot year has to confirm before the expansion is authorized.

Table 3.3 Planning assumptions of the financial analysis

Assumption	Value	Basis
Discount rate	10 % a year	hurdle rate of a small software house for an internal product investment
Horizon	Year 0 (build) + Year 1 (pilot, productization) + Years 2 to 5 (expansion)	useful life of the product before a major rewrite
Pilot implementation fee	150,000 THB, paid 30 % at LCA, 40 % at IOC, 30 % at go-live	fixed price for a custom LINE booking system of this size; the rest of the build is the software house's product investment
Plan prices	Free 0, Standard 690, Premium 990 THB a month; setup 2,500 THB	ceiling of 1,000 THB a month including hosting; Premium below the ceiling; setup covers 2 hours of configuration and training
Plan mix and average revenue	50 % Standard, 50 % Premium: 840 THB a month per paying shop	Premium carries the features restaurants ask for first (reminders, waitlist, reports)
Adoption	paying shops at year end 15, 40, 80, 130 in Years 2 to 5; Free-plan shops 20, 50, 90, 140; churn 10 % a year	about 500 addressable venues; 26 % share by Year 5; one reference customer in Year 2
Team cost rates	350 THB/h (Project Manager, System Analyst / Architect), 300 THB/h (Requirements Specifier / Designer, QA / Tester)	the software house's internal cost rates for its engineers, 2026
Team availability	6 to 16 hours a week per engineer in Year 0, 2 to 7.5 hours a week in Year 1 (Table 3.4)	part-time allocation alongside the software house's other work
Hosting	700 THB a month for the service plus 50 THB a month per shop, free or paying	one application server and database serving all shops; storage and messages grow per shop
Support and maintenance	16 hours a month base plus 0.5 hour a month per paying shop, at 300 THB/h	second-line support, LINE and gateway API changes, concert-night incidents
Sales and marketing	2,000 THB a month from Year 2	LINE ads, venue visits, demo events; the Free plan and the "Powered by CTRS" footer do the rest
Contingency reserve	10 % of the development cost of Year 0 and of the productization work of Year 1	known risks of Table 2.24; a management reserve is not budgeted
Payment gateway fees, LINE Official Account plan, door tablet	none for the software house	each restaurant holds its own merchant account and LINE plan and buys its own check-in device (about 9,000 THB)

3.3 Cost Analysis

3.3.1 Effort and labor cost

Effort is estimated top-down per phase from the planned hours a week of each engineer (Table 3.4): Year 0 builds Release 1.0 and 2.0 for the pilot restaurant in the eight iterations of the schedule (Section 2.8), Year 1 adds the productization of Release 2.1 (C4, T3). The engineers hold the RUP roles fixed in the project's Development Case; in Construction the Designer and the QA engineer also implement. Milestone 2 replaces this top-down figure by a bottom-up estimate per WBS task.

Table 3.4 Effort and labor cost by phase

Phase (iterations)	Year	Weeks	Project Manager (Boonnisa T.)	System Analyst (Sirapatr C.)	Requirements Specifier (Chawanakorn S.)	Quality Assurance (Watayut A.)	Hours
Inception (I1, I2)	0	4	32	40	32	24	128
Elaboration (E1, E2)	0	8	64	96	96	64	320
Construction Release 1.0 (C1, C2)	0	16	128	224	256	192	800
Transition Release 1.0 (T1)	0	5	40	40	50	60	190
Construction Release 2.0 (C3)	0	12	72	144	168	120	504
Transition Release 2.0 (T2)	0	5	30	30	40	50	150
Construction Release 2.1 productization (C4)	1	24	60	120	180	120	480
Transition Release 2.1 (T3)	1	4	8	12	20	20	60
Total hours		**78**	**434**	**706**	**842**	**650**	**2,632**
Rate (THB/h)			350	350	300	300	
Labor cost (THB)			**151,900**	**247,100**	**252,600**	**195,000**	**846,600**
of which Year 0 (build for the pilot restaurant)	0	50					674,600
of which Year 1 (productization)	1	28					172,000

3.3.2 Capital expenditure (CAPEX)

The service runs in the cloud and the engineers develop on the software house's existing laptops; capital expenditure is limited to the test devices for the LINE in-app browser and the door check-in (Table 3.5). The pilot restaurant buys its own check-in tablet.

Table 3.5 CAPEX

CAPEX	Unit price	Units	Year 0	Years 1 to 5	Total
Test devices: Android phone and 10-inch tablet for LIFF and check-in testing	15,000	1	15,000	-	15,000
Total CAPEX			**15,000**	-	**15,000**

3.3.3 Operating expenditure (OPEX)

Operating expenditure in Years 0 and 1 is the development labor of Table 3.4, the hosting and services used during the build and the pilot, and the company and legal set-up the subscription business needs; from Year 2 it is the running cost of the multi-tenant service, which scales with the number of shops (Table 3.6).

Table 3.6 OPEX

OPEX	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Total
Development labor (Table 3.4)	674,600	172,000	-	-	-	-	846,600
Development and staging hosting (12 months x 300 THB)	3,600	-	-	-	-	-	3,600
Production hosting for the pilot restaurant (Mar-Jul 2027, 5 months x 700 THB)	3,500	-	-	-	-	-	3,500
Domain name and TLS certificate (first year)	1,200	-	-	-	-	-	1,200
Payment gateway sandbox, LINE Developers, GitHub, Figma, PlantUML (free tiers)	-	-	-	-	-	-	0
Company registration, terms of service, privacy policy and data-processing agreement (legal)	-	15,000	-	-	-	-	15,000
Production hosting during the pilot year (12 months x 700 THB)	-	8,400	-	-	-	-	8,400
Support of the pilot restaurant (8 h/month x 300 THB)	-	28,800	-	-	-	-	28,800
Hosting of the service: base (700 THB/month) plus 50 THB/month per shop	-	-	19,200	45,900	86,400	140,400	291,900
Support and maintenance: 16 h/month base plus 0.5 h/month per paying shop, at 300 THB/h	-	-	71,100	107,100	165,600	246,600	590,400
Sales and marketing (2,000 THB/month)	-	-	24,000	24,000	24,000	24,000	96,000
Onboarding of new paid subscriptions (2 h each)	-	-	9,000	16,200	26,400	34,800	86,400
Total OPEX	**682,900**	**224,200**	**123,300**	**193,200**	**302,400**	**445,800**	**1,971,800**

3.3.4 Total cost

Table 3.7 Total cost

Cost	Details	Total (THB)
CAPEX	Test devices	15,000
OPEX, Year 0 build	Labor of the eight-iteration build, development and pilot hosting, domain	682,900
Contingency reserve, Year 0	10 % of the Year 0 cost, for the risks of Table 2.24	69,790
Project budget, Year 0		**767,690**
OPEX, Year 1 pilot and productization	Labor of C4 and T3, company and legal set-up, hosting and support of the pilot restaurant	224,200
Contingency reserve, Year 1	10 % of the productization work	18,700
Project budget, Year 1		**242,900**
Project budget, Years 0 and 1	development, pilot and productization	**1,010,590**
OPEX, Years 2 to 5	Operation of the subscription service: hosting, support, sales, onboarding	1,064,700
Total cost, Years 0 to 5		**2,075,290**

Record 8: Financial analysis, revenue, cost-benefit analysis and sensitivity

<table>
<tr><td>1. Artifact</td><td>Financial analysis Sections 3.4 to 3.7: revenue per year, cost-benefit analysis (NPV, ROI, payback, IRR, life-cycle cost), sensitivity with the partners' cash view, expansion gate and conclusion.</td></tr>
<tr><td>2. Member</td><td>Watayut Aiamanan</td></tr>
<tr><td>3. AI Tool</td><td>Claude Code (Claude Fable 5.1), continuing the session of Record 7.</td></tr>
<tr><td>4. Initial Prompt</td><td>[role] You are a project manager completing the financial analysis of a PMBOK business case.

[task] Quantify the benefits of CTRS from the ranked problems of the Business Case and the restaurant's volumes, and compute a five-year cost-benefit analysis at a 10 % discount rate on the cost model of Record 7 (all pasted with the prompt).

[format] A benefit table with a basis column; a discounted cash-flow table (Year 0 to Year 5) with NPV, ROI, discounted payback, IRR and life-cycle cost; a sensitivity table.</td></tr>
<tr><td>5. Initial Output</td><td>Six benefit lines (Manager time released, double-booking and dispute losses, slip losses, freed tables resold, fewer no-shows, tables sold to customers who no longer wait in chat) totalling 471,600 THB a year; NPV 762,824 THB, ROI 113 %, discounted payback 2.2 years, IRR 44 %. Excerpt:

<table><tr><th>Benefit</th><th>Basis</th><th>THB a year</th></tr><tr><td>Manager time released</td><td>3 hours x 48 nights x 350 THB</td><td>50,400</td></tr><tr><td>Freed tables resold</td><td>1 table x 48 nights x 3,500 THB</td><td>168,000</td></tr><tr><td>Double-booking refunds and disputes avoided</td><td>1 case x 48 nights x 3,500 THB</td><td>168,000</td></tr><tr><td>Slip losses avoided</td><td>0.5 % of table fees</td><td>25,400</td></tr></table></td></tr>
<tr><td>6. Review</td><td>Requirement satisfaction: NPV, ROI, payback and IRR present. PMBOK correctness: discounted cash flow with cumulative rows, undiscounted ROI stated as such.

Weaknesses found at the review: the return was the restaurant's benefit (Manager time, lost tables), which the software house never receives, and the volumes behind it were unvalidated; the result (NPV 762,824 THB, IRR 44 %) therefore overstated the case for the party that actually invests.</td></tr>
<tr><td>7. Revision</td><td>With the model re-framed (Record 7), the prompt was re-run with the task changed:

[task] Recompute the return from the software house's revenues only: the pilot fee, the subscriptions on the adoption plan and the setup fees; then show the sensitivity to the levers the software house controls (adoption plus or minus 30 %, prices minus 20 %, no pilot fee, stopping after the pilot year, the partners' cash view without the Year 0 and Year 1 labor) and propose written criteria for the expansion decision.

The base case became thin (NPV 18,971 THB, IRR 11 %, discounted payback 5.0 years), so the sensitivity table, the expansion gate with three criteria and a conclusion that recommends the project as a staged product</td></tr>
</table>

investment rather than as a one-off build were adopted.

8. Final Output

Part 2, Sections 3.4 to 3.7, as delivered below. Acceptable because the return now accrues to the party that invests, the thin base case is shown as it is, the levers and the downside (loss limited to the net investment of Years 0 and 1 if the gate is not passed) are explicit, and the decision the sponsor has to take is stated as gate criteria. Review Board: Revise; accepted after the revision.

3.4 Revenue Analysis

The software house earns the pilot implementation fee in Year 0, nothing in the pilot year, and from Year 2 the subscriptions of Table 3.1 on the paying shops of Table 3.2 plus a setup fee for every new paid subscription (Table 3.8). Free-plan shops earn nothing and cost hosting; they are the sales funnel.

Table 3.8 Revenue per year

Revenue	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Total
Pilot implementation fee (La Loy)	150,000	-	-	-	-	-	150,000
Subscriptions (840 THB a month per paying shop on average, 50 % Standard, 50 % Premium)	-	-	75,600	277,200	604,800	1,058,400	2,016,000
Setup fees (2,500 THB per new paid subscription)	-	-	37,500	67,500	110,000	145,000	360,000
Total revenue	**150,000**	**-**	**113,100**	**344,700**	**714,800**	**1,203,400**	**2,526,000**

3.5 Cost-Benefit Analysis

Table 3.9 brings the costs of Section 3.3 and the revenues of Section 3.4 together as a discounted cash flow; Table 3.10 derives from it the net present value (NPV), the return on investment (ROI, undiscounted, over Years 0 to 5), the discounted payback period, the internal rate of return (IRR) and the discounted life-cycle cost.

Table 3.9 Cost-benefit analysis

Item (discount rate 10 %)	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Total
Discount factor	1.0000	0.9091	0.8264	0.7513	0.6830	0.6209	
1. Costs							
1.1 CAPEX	15,000	-	-	-	-	-	15,000
1.2 OPEX	682,900	224,200	123,300	193,200	302,400	445,800	1,971,800
1.3 Contingency reserve	69,790	18,700	-	-	-	-	88,490
Total costs	767,690	242,900	123,300	193,200	302,400	445,800	2,075,290
Total discounted costs	767,690	220,818	101,901	145,154	206,543	276,807	1,718,913
Cumulative discounted costs	767,690	988,508	1,090,409	1,235,563	1,442,106	1,718,913	
2. Revenues							
2.1 Pilot implementation fee	150,000	-	-	-	-	-	150,000
2.2 Subscriptions	-	-	75,600	277,200	604,800	1,058,400	2,016,000
2.3 Setup fees	-	-	37,500	67,500	110,000	145,000	360,000
Total revenues	150,000	-	113,100	344,700	714,800	1,203,400	2,526,000

Item (discount rate 10 %)	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Total
Total discounted revenues	150,000	-	93,471	258,978	488,218	747,217	1,737,884
Cumulative discounted revenues	150,000	150,000	243,471	502,449	990,667	1,737,884	
3. Net cash flow							
Net cash flow (revenues - costs)	-617,690	-242,900	-10,200	151,500	412,400	757,600	450,710
Discounted net cash flow	-617,690	-220,818	-8,430	113,824	281,675	470,410	18,971
Cumulative discounted net cash flow	-617,690	-838,508	-846,938	-733,114	-451,439	18,971	

Table 3.10 Financial results

Measure	Value	Meaning
Net present value (NPV, 10 %)	**18,971 THB**	present value of revenues minus costs over Years 0 to 5; positive, but small against the 1,010,590 THB invested in Years 0 and 1
Return on investment (ROI, undiscounted, Years 0 to 5)	**22 %**	(total revenues - total costs) / total costs = (2,526,000 - 2,075,290) / 2,075,290
Discounted payback period	**5.0 years (during Year 5)**	cumulative discounted net cash flow turns positive
Internal rate of return (IRR)	**11 %**	discount rate at which NPV = 0; just above the 10 % hurdle
Life-cycle cost (discounted, Years 0 to 5)	**1,718,913 THB**	present value of all costs, development, pilot and operation
Break-even scale	**about 40 paying shops**	the year in which subscriptions and setup fees first exceed the operating cost and pay off the investment is Year 3 to 4 at the base adoption

Results. At the base-case assumptions the investment is recovered at the end of Year 5 with an NPV just above zero and an IRR just above the 10 % hurdle: the software house invests about 860,000 THB net in Years 0 and 1, the service breaks even on its operating cost at about 15 paying shops (Year 2), and the surplus of Years 4 and 5 (about 410,000 and 760,000 THB a year) pays the investment back. Within five years the case is therefore marginal; its value lies in Year 6 and later, when a service with 130 paying shops earns about 750,000 THB a year net without further development, and in the product the software house then owns.

3.6 Sensitivity and the Partners' Cash View

Table 3.11 recomputes the result for the levers the software house controls or must watch: adoption, prices, the pilot fee, the decision to stop after the pilot year, and the partners' cash position if they defer their own pay for this project.

Table 3.11 Sensitivity of the result

Case	Budget Years 0 and 1	Revenue Years 0 to 5	NPV (10 %)	ROI	Discounted payback	IRR
Base case (adoption as planned, pilot fee 150,000 THB, labor at the software house's cost rates)	1,010,590	2,526,000	18,971	22 %	5.0 years	11 %
Adoption 30 % higher (paying shops 20 / 52 / 104 / 169 at year end)	1,010,590	3,240,840	355,608	42 %	4.4 years	20 %

Case	Budget Years 0 and 1	Revenue Years 0 to 5	NPV (10 %)	ROI	Discounted payback	IRR
Adoption 30 % lower (paying shops 10 / 28 / 56 / 91)	1,010,590	1,808,660	-319,093	-3 %	beyond Year 5	-1 %
Prices 20 % lower (Standard 552, Premium 792, setup 2,000)	1,010,590	2,050,800	-298,606	-1 %	beyond Year 5	-1 %
No pilot implementation fee (La Loy as an unpaid design partner)	1,010,590	2,376,000	-131,029	14 %	beyond Year 5	6 %
Stop after the pilot year (expansion gate not passed): exposure of the software house	1,010,590	150,000	-838,508	-85 %	beyond Year 5	n/a
Partners' cash view: the four partners draw no salary for this project in Years 0 and 1	79,330	2,526,000	933,031	121 %	2.5 years	above 500 %

Adoption is the decisive lever: 30 % more shops turn the case clearly positive, 30 % fewer shops or a 20 % lower price turn it negative, and the 1,000 THB a month price ceiling leaves no room to compensate a slow uptake by charging more. The pilot fee matters because it is the only revenue before Year 2: without it the case does not pay back within the horizon. The exposure of the software house is bounded by the expansion gate: if the pilot does not confirm the plan, the loss is limited to the net investment of Years 0 and 1. From the partners' cash view the project needs about 80,000 THB of cash and returns it within the third year, which is why a four-partner software house can carry it as a product investment.

3.7 Conclusion

The project is financially justified as a product investment with a stage gate, not as a one-off custom build: the pilot implementation fee covers only a fifth of the build, and the return depends on selling the service to other restaurants at up to 990 THB a month. The team recommends that the software house authorizes Years 0 and 1 (build, pilot and productization, 1,010,590 THB with reserves, 860,590 THB net of the pilot fee) and decides the expansion at the end of the pilot year against three gate criteria: (1) La Loy meets the pilot success criteria of the charter and continues on a paid plan; (2) at least 10 restaurants are on the waiting list for the Free or a paid plan; (3) the productized Release 2.1 runs La Loy's concert nights with the per-shop hosting and support cost at or below the figures of Table 3.3. The financial analysis is re-run with the bottom-up cost estimate of Milestone 2 at the Lifecycle Architecture milestone and with the pilot data at the expansion gate.